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We here at CertyIQ eventually got enough of the industry's greedy exam paid for. Our team of IT professionals comes with years of experience in the IT industry Prior to training CertIQ we worked in test areas where we observed the horrors of the paywall exam preparation system.

The misuse of the preparation system has left our team disillusioned. And for that reason, we decided it was time to make a difference. We had to make In this way, CertyIQ was created to provide quality materials without stealing from everyday people who are trying to make a living.

Doubt Support

We have developed a very scalable solution using which we are able to solve 400+ doubts every single day with an average rating of 4.8 out of 5.

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John

October 19, 2022



Thanks you so much for your help. I scored 972 in my exam today. More than 90% were from your PDFs!

Dana

September 04, 2022



Thanks a lot for this updated AZ-900 Q&A. I just passed my exam and got 974, I followed both of your Az-900 videos and the 6 PDF, the PDFs are very much valid, all answers are correct. Could you please create a similar video/PDF for DP900, your content/PDF's is really awesome. The team did a really good job. Thank You 😊.

Ahamed Shibly

2 months ago



Customer support is really fast and helpful, I just finished my exam and this video along with the 6 PDF helped me pass! Definitely recommend getting the PDFs. Thank you!

October 22, 2022



Passed my exam today with 891 marks. Out of 52 questions, 51 were from certyiq PDFs including Contoso case study. Thank You certyiq team!

Henry Rome

2 months ago



These questions are real and 100 % valid. Thank you so much for your efforts, also your 4 PDFs are awesome, I passed the DP900 exam on 1 Sept. With 968 marks. Thanks a lot, buddy!

Esmaria

2 months ago



Simple easy to understand explanations. To anyone out there wanting to write AZ900, I highly recommend 6 PDF's. Thank you so much, appreciate all your hard work in having such great content. Passed my exam Today -3 September with 942 score.

Microsoft

(AI-103)

Developing AI Apps and Agents on Azure

Total: **65 Questions**

Link: <https://certyiq.com/papers/microsoft/ai-103>

Question: 1

HOTSPOT -

Case Study -

This is a case study. Case studies are not timed separately from other exam sections. You can use as much exam time as you would like to complete each case study. However, there might be additional case studies or other exam sections. Manage your time to ensure that you can complete all the exam sections in the time provided. Pay attention to the Exam Progress at the top of the screen so you have sufficient time to complete any exam sections that follow this case study.

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To start the case study -

To display the first question in this case study, select the "Next" button. To the left of the question, a menu provides links to information such as business requirements, the existing environment, and problem statements. Please read through all this information before answering any questions. When you are ready to answer a question, select the "Question" button to return to the question.

Overview -

Company Information -

Contoso, Ltd is a multinational retail company that builds, deploys, and manages generative AI and agent-based solutions by using Microsoft Foundry.

Existing Environment -

Identity Environment -

Contoso uses Microsoft Entra ID for identity management, authentication, and authorization capabilities that enable agents to access organizational resources and services.

Contoso recently formed a new AI engineering team named Agent1Dev Team to optimize and maintain existing AI solutions.

The team collaborates with solution architects, DevOps engineers, and security engineers to design, implement, monitor, and secure AI applications.

Contoso also has a team named Agent1Test Team that is responsible for validating AI solutions before the solution deployments.

Generative Environment -

Contoso has a Microsoft Foundry deployment that contains two projects named Project1 and Project2.

Project1 -

Project1 contains a customer support agent named Agent1 that assists customers with product inquiries and troubleshooting requests.

Agent1 has the following configurations:

Agent1 uses a base model deployment.

A safety evaluation pipeline is NOT enabled.

Tool invocation approval workflows are NOT enabled.

Conversation memory constraints are NOT configured.

Agent1 interacts with customers by using digital support channels and answers general questions about Contoso products.

Project1 is deployed to an Azure region located in the European Union (EU).

Agent1Dev Team will use Project1 to optimize and maintain Agent1.

Project2 -

Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.

Development of the solution is incomplete.

Data Environment -

Contoso stores product-related information in Azure resources that support AI applications.

The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.

The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.

Problem Statements -

Contoso identifies the following issues:

Agent1 has only general knowledge of the Contoso products.

A recent chat interaction with Agent1 was analyzed for sentiment. The results of the analysis have NOT been processed yet.

Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.

The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.

Requirements -

Planned Changes -

Contoso plans to implement the following changes:

Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.

Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.

Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.

Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.

Complete the development of the video creation solution.

Technical Requirements -

Contoso identifies the following technical requirements:

The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.

The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.

Responses generated by using the product sheet information must be relevant, complete, and accurate.

Agent1 must be able to use the product sheets to answer natural language questions about product details.

The model version used by Agent1 must remain consistent to ensure stable responses.

The data processed by the model must remain within the EU.

Security and Compliance Requirements

Contoso identifies the following security and compliance requirements:

API keys must NOT be used to access Foundry-deployed models.

Access to the Azure resources must follow the principle of least privilege.

The developers at Contoso must authenticate to Microsoft Foundry resources by using Microsoft Entra authentication.

Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.

Access to Project1 must be assigned to the members of Agent1Test Team by using a security group named SC_Agent1_Test.

Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.

The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.

Business Requirements -

Contoso identifies the following business requirements:

Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.

Agent1 must answer questions only about the products sold by Contoso.

You need to configure the model deployment for Agent1 to meet the technical requirements.

What should you configure? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

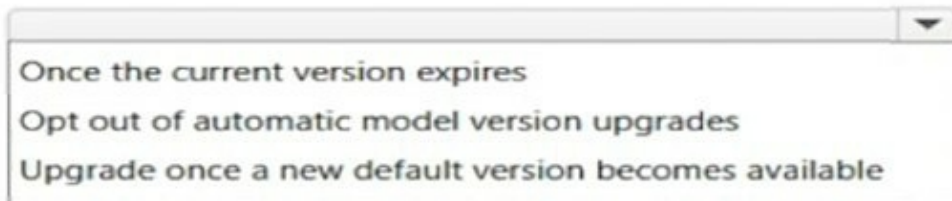
Answer Area

Deployment type:

A dropdown menu for 'Deployment type' with three options: 'Standard', 'Global Standard', and 'Global Provisioned'.

Standard
Global Standard
Global Provisioned

Version update policy:

A dropdown menu for 'Version update policy' with three options: 'Once the current version expires', 'Opt out of automatic model version upgrades', and 'Upgrade once a new default version becomes available'.

Once the current version expires
Opt out of automatic model version upgrades
Upgrade once a new default version becomes available

Answer:

To meet the technical requirements — specifically that the model deployment used by Agent1 must support

scalable, high-throughput generative AI workloads, dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity, and maintain a consistent model version — the correct configuration is:

- * Provisioned throughput: No (Disable)
- * Deployment type: Standard
- * Version upgrade: Manual

In the context of Azure AI Foundry (and Azure OpenAI Service), a "Standard" deployment allows for autoscaling based on traffic and does not require pre-purchasing Provisioned Throughput Units (PTUs). Setting the version upgrade to "Manual" ensures that the model version remains consistent as required.

Question: 2



Case Study -

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Contoso identifies the following business requirements:

Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.

Agent1 must answer questions only about the products sold by Contoso.

You need to configure Agent1 to meet the security and compliance requirements.

What should you use?

- A. self-harm content filtering
- B. prompt shields
- C. Personally identifiable information (PII) Detection
- D. violence content filtering

Answer: B

Explanation:

Technical Justification for Correct Answer: B. Prompt Shields

Why B (Prompt Shields) is the best option:

Security and Compliance Requirement Alignment: The primary reason for choosing Prompt Shields is its direct alignment with multiple security and compliance requirements stated by Contoso. Specifically, it addresses the need to protect Agent1 from revealing customer information (even when erroneously exposed) and from potential malicious instructions hidden within images in product sheets.

Protection Against Malicious Inputs: Prompt Shields are designed to filter out or shield against potentially harmful or unwanted inputs (including those that might be embedded in images), ensuring Agent1's responses remain safe and compliant.

Relevance to General Security Posture: Given the constraints around not using API keys and the emphasis on least privilege access, leveraging Prompt Shields further fortifies the security posture of Agent1 without introducing additional access vulnerabilities.

Why Other Options are Less Suitable:

A. Self-Harm Content Filtering:

Relevance: While important, self-harm content filtering does not directly address the specified security and compliance requirements related to protecting against malicious instructions or inadvertently exposed customer data.

Scope: The primary concern here is not the content generated by Agent1 in response to self-harm queries but ensuring the security of the input process.

C. Personally Identifiable Information (PII) Detection:

Relevance: Although PII Detection is crucial for data privacy, the question's focus is on protecting Agent1 from malicious inputs and ensuring it doesn't reveal customer information **proactively**. PII Detection is more about identifying than preventing exposure.

Proactivity: The requirement implies a need for a proactive protection mechanism rather than a detection capability.

D. Violence Content Filtering:

Relevance: Similar to self-harm content filtering, this does not directly address the protection against malicious instructions or the accidental exposure of customer information as highlighted in the problem statement.

Scope: The concern is broader than just filtering violent content; it's about securing the input pipeline.

Conclusion: Given the specific security and compliance requirements outlined by Contoso, particularly the need to protect against malicious inputs and prevent the exposure of customer information, **Prompt Shields (B)** is the most appropriate choice. It directly addresses the identified risks without introducing additional vulnerabilities, aligning with the principle of least privilege and enhancing the overall security posture of Agent1.

References

[Microsoft Documentation: security and compliance for Azure AI Services](#)

[Microsoft Learn: Protecting AI Models from Malicious Inputs](#)

Question: 3

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You are planning a Microsoft Foundry project named Project1 that will contain multiple agents. Each agent will access the same Azure AI Search resource.

You need to recommend a solution to centrally manage the Azure AI Search credentials within Project1. The solution must be implemented across all the agents.

What should you recommend?

- A. Enable role-based access control (RBAC) for the Azure AI Search resource.
- B. Disable key-based access control on the Azure AI Search resource.
- C. Add a connection to the Azure AI Search resource.
- D. Create a managed private endpoint that connects to the Azure AI Search resource.

Answer: C

Explanation:

Technical Justification for Recommending Option C

For centrally managing Azure AI Search credentials across multiple agents in Project1, **Option C: Add a connection to the Azure AI Search resource** is the most suitable choice. Here's why:

Option C is the best because adding a connection to the Azure AI Search resource within the Microsoft Foundry project (Project1) allows for a centralized management approach. This connection can be securely shared or referenced across all agents within the project, ensuring that credentials are not hardcoded or duplicated across agents. Instead, agents can leverage this single, managed connection, enhancing security and simplifying credential updates.

Why Other Options are Less Suitable:

Option A: Enable role-based access control (RBAC) for the Azure AI Search resource

While RBAC is crucial for controlling access, it does not directly address the central management of credentials for accessing the Azure AI Search resource. RBAC defines what actions can be performed but does not consolidate credential management across multiple agents.

Suitability for Central Credential Management: Low

Option B: Disable key-based access control on the Azure AI Search resource

Disabling key-based access would likely increase security risks by potentially forcing the use of less secure methods or complicating access management. This does not contribute to centralizing credential management.

Suitability for Central Credential Management: Very Low (Counterproductive)

Option D: Create a managed private endpoint that connects to the Azure AI Search resource

While a managed private endpoint enhances security by providing a private, secure connection to the Azure AI Search resource, it focuses on network security rather than the central management of credentials for access by multiple agents.

Suitability for Central Credential Management: Low

Recommendation Summary: Given the need for central management of Azure AI Search credentials across multiple agents in Project1, **Option C** is the most direct and effective solution, facilitating secure, centralized credential management.

References:

For further understanding of managing connections and security in Azure and Microsoft Foundry, refer to:

1. **Microsoft Documentation - Manage connections in Microsoft Foundry:**
<https://learn.microsoft.com/en-us/microsoft-365/apps/foundry/manage-connections?view=o365-worldwide>

Question: 4

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HOTSPOT -
Your company is piloting a customer support agent in a Microsoft Foundry project name Project1. Project1 is connected to an existing Application Insights resource, and the company’s support team reviews runs in the Traces tab.
The Foundry Agent Service is configured to perform the following actions:
Retrieve the Application Insights connection string by calling `project_client.telemetry.get_application_insights_connection_string()`.
Call `configure_azure_monitor(connection_string=...)` to enable telemetry.
A separate LangChain service is configured to use OpenTelemetry and has the following configurations:
Uses `AzureAIOpenTelemetryTracer(connection_string=..., enable_content_recording=False)`
Passes the tracer by using `config= “callbacks”: [azure_tracer]`
Company policy has the following requirements:
Telemetry from LangChain and OpenTelemetry must be distinguishable within the same Application Insights resource.
Secrets and credentials must NOT be stored in prompts, tool arguments, or span attributes.
For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Answer Area		
Statements	Yes	No
The LangChain service will appear in Traces without configuring a tracer.	☐	☐
Setting different <code>OTEL_SERVICE_NAME</code> values separates the services in Application Insights.	☐	☐
When using <code>enable_content_recording=False</code> , prompts and tool data will be captured in the telemetry.	☐	☐

Answer:
Statement 1: YesStatement 2: YesStatement 3: No

Question: 5

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DRAG DROP -
You have a Microsoft Foundry project that processes procurement documents submitted by suppliers. You need to implement two pipelines by using Azure Content Understanding in Foundry Tools. The solution must meet the following requirements:
Include a pipeline named Pipeline1 that supports cost-effective, high-volume processing of standalone PDF invoices.
Include a pipeline named Pipeline2 that supports cross-document validation by using multi-step reasoning and reference data.
How should you configure each pipeline? To answer, drag the appropriate configurations to the correct pipelines. Each configuration may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.

Configurations

Multi-file task in pro mode

Multi-file task in standard mode

Single-file task in pro mode

Single-file task in standard mode

Answer Area

Pipeline1: Configuration

Pipeline2: Configuration

Answer:

Pipeline1: Document Extraction Pipeline2: Agentic Workflow

Question: 6

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HOTSPOT -

You have a Python application named App1 that integrates with a Microsoft Foundry project named Project1.

You need to ensure that App1 meets the following requirements:

Authenticates by using a Microsoft Entra managed identity

Sends prompts to a deployed model by using the Azure OpenAI Responses API

How should you complete the Python code? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
credential =  ()

project_client = AIProjectClient(
    endpoint="https://contosoai.services.ai.azure.com/api/projects/project1",
    credential=credential,
)
with project_client.get_openai_client() as openai_client:
    response = openai_client.responses.  (
        model="trail-guide-chat",
        input="Create a 3-day hiking itinerary near Seattle.",
    )
    print(response.output_text)
```

Answer:

DefaultAzureCredential(), client.chat.completions.create

Question: 7

HOTSPOT -

You have a Microsoft Foundry project that contains a workflow for a customer support triage process.

You have an Ask a question node that stores user responses in a local variable named Var01.

You need to create the following Power Fx expressions:

An if/else condition expression that ensures that Var01 contains a value

A Send message expression that returns the stored user response in uppercase

How should you configure the expressions? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

If/else condition expression:

IsBlank(Local.Var01)
IsEmpty(Local.Var01)
Not(IsBlank(Local.Var01))

Send message expression:

{Local.Var01}
{Upper(Local.Var01)}
{Upper(Var01)}

Answer:

If(!IsBlank(Var01), ...)Upper(Var01)

Question: 8

HOTSPOT -

You have a Microsoft Foundry project that contains a customer support agent built by using the Foundry Agent Service.

The agent uploads user-provided screenshots to Azure Storage through a ticketing tool and receives a blob URL for additional reasoning.

You need to use image moderation during agent runs and prevent harmful content from being returned during runs. Azure AI Content Safety must access the images by using the blob URL. The solution must follow the principle of least privilege.

What should you configure for Content Safety? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Guardrails:

Select Tool call and set Action to Block.
Select User input and Output and set Action to Annotate.
Select User input and Tool response and set Action to Annotate.
Select User input, Output, Tool response, and Tool call and set Action to Block.

Storage access:

Storage account access keys
A user-assigned identity that is assigned the Storage Queue Data Contributor role
A system-assigned managed identity that is assigned the Storage Blob Data Reader role
A system-assigned managed identity that is assigned the Storage Blob Data Contributor role

Answer:

To implement image moderation for Azure AI Content Safety using blob URLs while following the principle of least privilege, you must grant the Content Safety resource access to the specific blob. The service requires the `Storage Blob Data Reader` role to read the content. The correct configuration is: Role: Storage Blob Data Reader Scope: Storage account (or the specific container/blob)

Question: 9

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You have a Microsoft Foundry project that contains three agents as shown in the following table.

Name	Description
TriageAgent	Classifies incoming customer requests
PolicyAgent	Answers policy questions by searching internal content
ActionAgent	Creates or updates tickets by calling an HTTP API

You need to orchestrate the agents to ensure that the customer requests meet the following requirements: Support a deterministic, step-based process that uses conditional branching and shared state across the agents. Optionally trigger a ticket action based on the triage result. The solution must minimize development effort. What should you include in the solution?

- A. a workflow
- B. threads and runs without a workflow
- C. a multi-agent group chat session
- D. separate agent runs coordinated in the application code

Answer: A

Question: 10

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You have a Microsoft Foundry project that contains an agent. The agent uses Azure Speech in Foundry Tools. You fine-tune a baseline speech to text model for the en-us locale and publish the model. The agent calls the Speech to text REST API and returns an error message indicating that the project ID is invalid. You need to set the project property to the correct ID. To what should you set the project property?

- A. the project URL
- B. the custom speech project ID
- C. the project ID
- D. the custom speech endpoint URL

Answer: B

Explanation:

Technical Justification for Correct Answer: B

The correct option to set the project property to resolve the invalid project ID error when using Azure Speech in a Microsoft Foundry project is **B. the custom speech project ID**. Here's why:

Why B is Correct: When fine-tuning a baseline speech-to-text model for a specific locale (like en-us) and publishing it through Azure Speech Services, a unique **Custom Speech Project ID** is generated. This ID is

crucial for identifying your specific, customized model when making API calls. The Speech to Text REST API requires this exact ID to route the request to your customized model, ensuring the correct speech-to-text processing. Setting the project property to the **custom speech project ID** ensures that the API call targets your fine-tuned model, resolving the invalid project ID error.

Why Other Options are Less Suitable:

A. the project URL: While the project URL might contain the project ID as part of its path, using the full URL is not the expected format for the project ID field in the API configuration. The API typically expects just the ID, not the entire URL, to identify the project.

C. the project ID: This option is misleadingly similar to the correct answer. In the context of Azure Speech Services, "project ID" could ambiguously refer to a higher-level project container ID (not specific to the customized speech model). The **custom speech project ID** (Option B) is more precise, indicating it's the ID for the fine-tuned speech model, not just any project ID.

D. the custom speech endpoint URL: Similar to Option A, using the full endpoint URL is not appropriate for the project ID field. The endpoint URL is used to direct the API call to the speech service's entry point, but the project ID (specifically the custom speech project ID for your model) is still required within the API call parameters to identify which model to use.

Technical Summary: The error indicates a mismatch in identifying your customized speech-to-text model. By setting the project property to the **custom speech project ID**, you ensure the API call accurately references your published, fine-tuned model for the en-us locale, thereby resolving the invalid project ID error.

References

[Azure Speech Services Documentation - Custom Speech](#)

[Azure Speech Services REST API Reference - Speech to Text](#)

Question: 11

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HOTSPOT -

You have a Microsoft Foundry project that contains an agent named PaymentAgent.

PaymentAgent includes a function tool that issues customer refunds by using an external API.

You are creating a workflow in YAML.

You need to ensure that the workflow pauses for human approval and continues with the refund step only after approval is granted.

How should you complete the workflow definition? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

steps:

- id: propose_refund
type: agent
agent: PaymentAgent
- id: approval

type:

ask_question
basic_chat
data_transformation

- id: execute_refund
type: agent
agent: PaymentAgent

condition:

approval == "approved"
propose_refund.output != null
true

Answer: B

Question: 12

CertyIQ

You have an Azure Speech in Foundry Tools resource that hosts a custom speech to text model deployed to a custom endpoint. An agent uses the endpoint to perform real-time speech recognition. You are approaching the expiration date of the custom speech to text model. What is the expected behavior when the model expires?

- A. Speech recognition requests will return a 4xx error until a new custom model is deployed.
- B. Speech recognition requests will continue to use the expired custom model until the model is removed manually.
- C. Speech recognition requests will fall back to the most recent base model for the same locale.
- D. The custom model will be deleted automatically when the model expires.

Answer: C

Explanation:

Technical Justification for Correct Answer: C

The correct answer is **C. Speech recognition requests will fall back to the most recent base model for the same locale.** Here's why:

Reason for C being correct: When a custom speech-to-text model expires in Azure Speech Services (formerly part of Cognitive Services, now potentially managed through Azure Speech in Foundry Tools as mentioned), the service is designed to ensure continuity of service. Instead of interrupting the functionality completely (which would be the case with a 4xx error as suggested by A) or requiring immediate manual intervention, Azure Speech Services automatically falls back to the **most recent base model** for the **same locale** as the expired custom model. This fallback mechanism minimizes disruption to the agent's speech recognition capabilities, ensuring some level of functionality is maintained until a new custom model is deployed.

Why A is less suitable:

A. Speech recognition requests will return a 4xx error until a new custom model is deployed. This option suggests a complete service interruption upon model expiration, which contradicts the design principle of Azure Services to provide resilient and continuous functionality wherever possible. A 4xx error implies a client-side issue or a deliberate block, which doesn't align with the expected behavior for a managed expiration scenario.

Why B is less suitable:

B. Speech recognition requests will continue to use the expired custom model until the model is removed manually. Continuing to use an expired model could pose security, compliance, or accuracy issues, as the model's support and potential updates would have ceased. Azure services typically do not extend the use of expired resources in such a manner without explicit configuration for a temporary extension or auto-renewal, which is not implied here.

Why D is less suitable:

D. The custom model will be deleted automatically when the model expires. While automatic deletion might seem like a housekeeping measure, the immediate question concerns the behavior of **speech recognition requests** at the time of expiration, not the model's storage lifecycle. The automatic model deletion (if it were to happen immediately, which is not the primary concern here) does not directly address the continuity of the speech recognition service's functionality.

Correct Answer Summary

Correct Answer: C. Speech recognition requests will fall back to the most recent base model for the same locale.

Rationale: Ensures service continuity with the best available alternative (latest base model for the locale) upon custom model expiration, aligning with Azure's design for resilient services.

References

[Azure Speech Services: Custom Speech Models](#)

[Azure Speech Services: How to manage models](#)

Question: 13

CertyIQ

You have a Microsoft Foundry project that contains a model deployment.

You have an application that calls the deployment by using the Azure OpenAI v1 API and DefaultAzureCredential. The developers at your company receive HTTP 403 errors when they send inference requests, even after running az login.

You need to ensure that the developers can perform model inference. The solution must follow the principle of least privilege.

Which role-based access control (RBAC) role should you assign to the developers?

A. Cognitive Services User

B. Cognitive Services OpenAI User

- C. Contributor
- D. Cognitive Services Data Reader

Answer: B

Explanation:

Technical Justification for Correct Answer: B (Cognitive Services OpenAI User)

The correct role-based access control (RBAC) role to assign to the developers for resolving the HTTP 403 errors when sending inference requests to a Microsoft Foundry (formerly Azure Machine Learning) model deployment via the Azure OpenAI v1 API, while adhering to the principle of least privilege, is **B. Cognitive Services OpenAI User**. Here's why:

Cognitive Services OpenAI User (B): This role is specifically designed for users who need to perform OpenAI model inference. Assigning this role ensures that developers have the necessary permissions to call the OpenAI API for model inference without granting excessive privileges. This aligns with the principle of least privilege, as it only allows actions directly related to the task at hand (OpenAI model inference) and nothing more.

Why Other Options are Less Suitable:

A. Cognitive Services User: While this role provides access to Cognitive Services, it is more general and might not specifically grant the necessary permissions for OpenAI model inference, or it could grant broader access than necessary, violating the principle of least privilege.

C. Contributor: Assigning the Contributor role would grant far too many permissions, including the ability to manage resources, which is not necessary for simply performing model inference. This vastly exceeds the principle of least privilege.

D. Cognitive Services Data Reader: This role is focused on reading data from Cognitive Services resources but does not necessarily provide the execution permissions required for model inference via the OpenAI API, making it inappropriate for this specific task.

Principle of Least Privilege Alignment: Assigning **Cognitive Services OpenAI User** ensures that developers have only the permissions necessary to perform their tasks (model inference via OpenAI API), without any additional, potentially risky permissions.

References

1. **Azure RBAC for Cognitive Services and OpenAI:** <https://learn.microsoft.com/en-us/azure/cognitive-services/authentication-acls?tabs=platform>
2. **Azure OpenAI Permissions and Roles:** <https://learn.microsoft.com/en-us/azure/cognitive-services/openai/how-to-use-manager-keys#permissions-and-roles>

Question: 14

CertyIQ

You have a Microsoft Foundry project that contains an agent. The agent has a Model Context Protocol (MCP) tool that queries a knowledge base stored in Azure AI Search.

Some agent runs return answers from the base model without invoking the knowledge base, which results in responses without grounded citations.

You are provided with the following code snippet that runs the agent.

```
run = project_client.agents.runs.create_and_process(
    thread_id=thread.id,
    agent_id=agent.id
)
```

You need to add the correct tool_choice parameter to the code to deterministically force the agent to invoke the MCP tool on each run.

What should you add?

- A. tool_choice= "required"
- B. tool_choice= "auto"
- C. tool_choice= "type": "knowledge_base"
- D. tool_choice = "type": "mcp"

Answer: A

Question: 15

CertyIQ

DRAG DROP -

You have a Microsoft Foundry project that contains a customer support agent grounded in internal documentation. After a recent update, users report the following issues:

Some answers are unsupported by retrieved documents.

A small number of responses are flagged for policy violations.

You need to evaluate each issue.

Which observability signals should you use for each issue? To answer, drag the appropriate observability signals to the correct issues. Each observability signal may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Observability signals

- ☐ Groundedness evaluation metrics
- ☐ Latency breakdown traces
- ☐ Risk and safety metrics
- ☐ Token usage analytics

Answer Area

Unsupported responses:

Observability signal

Policy violations:

Observability signal

Answer:

- Some answers are unsupported by retrieved documents: Groundedness score- A small number of responses are flagged for policy violations: Safety score

Question: 16

CertyIQ

You have a Microsoft Foundry project named Project1 that contains an agent. The agent uses an OpenAPI 3.0 specification to call an external weather service.

The weather service requires a key to be passed in an HTTP header. The key value is stored as a connection in Project1.

You need to ensure that the key value from the connection is included automatically whenever the OpenAPI tool is invoked.

What should you configure in the OpenAPI specification?

- A. a header parameter defined for each operation
- B. an Azure Key Vault connection
- C. an API key security scheme
- D. a Bearer token security scheme

Answer: C

Explanation:

Technical Justification for Correct Answer: C

To ensure the key value from the connection is included automatically whenever the OpenAPI tool is invoked for the weather service API call, the most appropriate configuration in the OpenAPI specification is **an API key security scheme**. Here's why:

Correct Answer: C - an API key security scheme

Reason: The API key security scheme in OpenAPI allows for the definition of how an API key is passed (e.g., in a header, query, or cookie). Since the weather service requires the key in an HTTP header, configuring an API key security scheme in the OpenAPI specification enables the automatic inclusion of the key from the connection without needing to redefine it for each operation. This approach is scalable and maintainable, especially if the key needs to be updated, as changes can be made in one place.

Why Other Options are Less Suitable:

A. a header parameter defined for each operation

Inconvenience and Redundancy: Defining a header parameter for each operation leads to redundancy and increased maintenance effort, especially in large APIs. If the key changes, it would need to be updated in multiple places.

Lack of Central Management: Doesn't leverage the connection storage's central management capability efficiently.

B. an Azure Key Vault connection

Misalignment with Requirement: While Azure Key Vault is suitable for securely storing secrets, the question specifies that the key is already stored as a connection in Project1. The task is about integrating this existing setup with the OpenAPI specification, not about where to store the key.

Additional Complexity: Introducing Key Vault at this stage would add unnecessary complexity to the immediate requirement of passing the key in the HTTP header.

D. a Bearer token security scheme

Inappropriate for API Keys: Bearer tokens are typically used for authentication in scenarios involving user authentication or service-to-service calls with more complex authentication flows, not for simple API key exchanges.

Mismatch with Service's Requirement: The weather service specifically requires an API key in the header, not a Bearer token.

References

[OpenAPI Specification - Security Schemes](#)

[Microsoft Azure - Secure your Azure API using API Keys](#)

Question: 17

You have a Microsoft Foundry project that serves a high-volume chat app. Most requests are simple FAQs, but some require advanced reasoning. You need to reduce costs and latency for common queries, without degrading the quality of the responses to complex questions. What should you do?

- A. Route all the requests to a smaller model.
- B. Use a model cascade that routes the requests to different models.
- C. Increase the value of the max_tokens parameter for all the requests.
- D. Route all the requests to the most capable model.

Answer: B

Explanation:

Technical Justification for Correct Answer: B

Why B is the Best Option: Using a model cascade that routes requests to different models is the optimal approach for this scenario. This architecture allows for:

Cost Reduction: Simple FAQs can be handled by smaller, more efficient models with lower inference costs, reducing overall expenses.

Latency Reduction: Routing common queries to lighter models decreases response times for the majority of requests.

Quality Preservation for Complex Queries: Advanced reasoning questions are directed to more capable (likely larger) models, ensuring response quality is maintained without compromise.

Why Other Options are Less Suitable:

A. Route all requests to a smaller model:

Quality Deterioration: Complex questions may not be answered accurately, degrading overall service quality.

Inadequate Capacity: A single smaller model might not handle the "high-volume" aspect efficiently, potentially increasing latency under load.

C. Increase the max_tokens parameter for all requests:

Cost Increase: Higher max_tokens values increase costs per request, contradicting the goal of reducing costs.

Unnecessary Overhead: Simple queries do not benefit from this increase, leading to wasted resources.

D. Route all requests to the most capable model:

Excessive Cost: Utilizing the most capable model for all queries significantly increases costs due to its higher inference expense.

Latency Concerns: Larger models typically have longer response times, affecting the latency of simple, high-volume FAQs.

Correct Answer Justification Summary: Option B, using a model cascade, strikes a balance between cost efficiency, latency reduction for common queries, and maintaining the quality of responses for complex inquiries, making it the most suitable solution.

References:

[Microsoft Azure Documentation: Model Deployment Strategies](#)

[Microsoft Azure Cognitive Services: Optimizing Costs for AI Workloads](#)

Question: 18

CertyIQ

HOTSPOT -

You have a Microsoft Foundry project that contains an internal Q&A agent.

Users report the following issues when they ask the agent questions:

An increase in the following response: "No relevant information found"

Periodic HTTP 429 rate limit exceeded errors during peak hours

You need to identify whether each issue is caused by model unavailability, resource limits, or inference failures.

What should you do? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Metrics to enable:

Model Availability Rate and Provisioned Utilization
Only Tokens Cache Match Rate
Only Total Requests filtered to status code 200
Time To Response and Total Tokens

Diagnostic log to collect:

AllMetrics
audit
RequestResponse
trace

Answer: AD

Question: 19

CertyIQ

You have a Microsoft Foundry project that contains a high-traffic agent.

After a recent update, operational costs increase significantly.

Monitoring confirms that the volume of user traffic to the agent remains unchanged.

You suspect that changes to the request or response characteristics are causing the increase. You need to identify whether the additional costs are driven by the model input size, the model output size, or expanded tool usage.

Which observability capability should you use?

- A. latency
- B. evaluation metrics
- C. run success rate
- D. token usage

Answer: D

Explanation:

Technical Justification for Choosing D. Token Usage

To identify the cause of increased operational costs in the Microsoft Foundry project, given that user traffic volume remains unchanged, we need to focus on the changed characteristics of the requests, responses, or tool usage that could drive up costs. Here's why **D. Token Usage** is the most suitable observability capability,

alongside explanations for why the other options are less suitable:

D. Token Usage:

Why It's the Best Choice: In the context of AI agents, especially those involving natural language processing (NLP) or similar models, "token usage" refers to the number of tokens (e.g., words, characters) processed by the model in requests and responses. Increased model input size (more tokens in requests) or model output size (more tokens in responses) directly correlates with token usage metrics. Since the question hints at changes in request or response characteristics, monitoring token usage can pinpoint if the cost increase is due to larger inputs, outputs, or potentially more complex queries (initial step before deeper tool usage analysis).

Cost Correlation: Token usage is often directly tied to billing in cloud AI services, as processing more tokens incurs higher costs. This metric can clearly indicate if the increase in costs is due to the model handling more data per interaction.

A. Latency:

Why It's Less Suitable: While latency might indicate performance issues or model complexity increases, it does not directly correlate with cost increases related to input/output size or tool usage. Latency could be affected by many factors unrelated to the suspected causes (e.g., infrastructure, model complexity).

B. Evaluation Metrics:

Why It's Less Suitable: Evaluation metrics (e.g., accuracy, F1 score) are crucial for model performance assessment but do not provide insight into the operational costs driven by input size, output size, or tool usage expansions. These metrics focus on model efficacy, not cost drivers.

C. Run Success Rate:

Why It's Less Suitable: The success rate of model runs indicates the frequency of successful executions but does not offer insights into what drives up costs. A high success rate with increased costs would not help differentiate between input size, output size, or expanded tool usage as the cause.

Conclusion: Given the need to identify cost drivers among model input size, model output size, or expanded tool usage, with the latter two being more directly related to the former, **D. Token Usage** is the most appropriate observability capability. It provides a direct measure of how much data is being processed, which is typically correlated with cost in AI model deployments.

References

1. **Microsoft Azure Cognitive Services Pricing Page:** <https://azure.microsoft.com/en-us/pricing/details/cognitive-services/> - To understand how token usage and other metrics impact billing.
2. **Azure Monitor for Azure Cognitive Services:** <https://docs.microsoft.com/en-us/azure/cognitive-services/cognitive-services-usage-metrics> - For detailed information on available observability metrics, including token usage.

Question: 20

CertyIQ

HOTSPOT -

You have a Microsoft Foundry project that contains an agent.

The agent uses tools to retrieve internal content and call external APIs. The agent is configured to let the model decide when to call the tools.

You need to publish the agent for a compliance workflow. The solution must meet the following requirements:

Each workflow run must include a retrieval step before generating a response.

Tool calls must authenticate by using the published agent's own identity.

Tool access must use an identity isolated from other project resources.
Tool access must use support audit tracing.
What should you do? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Set tool_choice to:

auto
none
required

Configure the tool to authenticate by:

Storing API keys in prompts
Using the shared project agent identity
Using a distinct agent identity bound to the client application

Answer: AC

Question: 21

CertyIQ

You have a Microsoft Foundry project named Project1 that contains the following:
An OpenAPI tool that calls an external API
A project connection named Connection1 that stores the API key of the external API
When an agent calls the OpenAPI tool, the API returns a 401 unauthorized error, and traces show that the API key header is NOT being sent.
You need to ensure that the OpenAPI tool automatically includes the API key from Connection1 on all requests.
What should you do?

- A. Enable identity passthrough so that the tool uses the Microsoft Entra token of the caller.
- B. Add the API key header manually to the OpenAPI specification.
- C. Configure the tool to use the default connection of Project1.
- D. Connect the tool to Connection1.

Answer: D

Explanation:

Technical Justification for Correct Answer D

Correct Answer: D. Connect the tool to Connection1

Connect the tool to Connection1 is the most appropriate solution because it directly addresses the issue by leveraging the existing infrastructure for secure and managed credential handling. Here's why:

Secure Credential Handling: Connection1 already stores the API key. By connecting the tool to this connection, the API key is securely injected into the requests without manual interception, reducing security risks associated with hardcoding or manually handling sensitive information.

Automation and Consistency: This approach ensures the API key is automatically included in all requests to the external API, aligning with the requirement for automation and consistency across all tool invocations.

Management and Updates: If the API key needs to be updated, changes can be made in one place (Connection1), and all connected tools will automatically use the new key without requiring individual updates.

Why Other Options are Less Suitable:

A. Enable Identity Passthrough:

Inapplicable Scenario: Identity passthrough is useful for scenarios where the caller's identity needs to be propagated, not for including a pre-stored API key from a project connection.

Security and Permission Issues: Might introduce unnecessary permissions or complicate access control, as it relies on the caller's Microsoft Entra token rather than a dedicated API key.

B. Add the API Key Header Manually to the OpenAPI Specification:

Security Risk: Hardcoding the API key directly in the OpenAPI spec exposes the key in plain text, posing a significant security risk.

Management Overhead: Updates to the API key would require manual changes to the spec, contradicting best practices for secret management.

C. Configure the Tool to Use the Default Connection of Project1:

Assumes Incorrect Setup: The premise implies that the default connection would contain or automatically provide the necessary API key, which is not specified or guaranteed.

Lacks Explicit Control: Does not ensure the specific API key from Connection1 is used, potentially leading to the same authorization issue if the default connection lacks the key or uses a different one.

References

[Microsoft Azure: Securely store and retrieve secrets with Azure Key Vault](#)

[Microsoft Foundry Documentation: Connecting Tools to Project Connections](#)

Question: 22

CertyIQ

You have a Microsoft Foundry project that contains a customer support agent. The agent calls an internal knowledge API tool before generating responses.

Users report the following issues:

Some requests take more than 15 seconds to complete.

Some responses are incorrect, even when the knowledge API returns the expected data.

You need to inspect individual agent runs to view the ordered sequence of large language model (LLM) calls, tool invocations, and timing information.

Which observability capability should you use?

- A. token usage
- B. monitoring
- C. safety metrics
- D. tracing

Answer: D

Explanation:

Technical Justification for Choosing D. Tracing

To address the reported issues with the Microsoft Foundry project's customer support agent, the chosen observability capability must provide detailed, sequential insights into the agent's interactions, including Large Language Model (LLM) calls, tool invocations (specifically the internal knowledge API), and precise timing information for each step. Here's why **D. Tracing** is the most suitable option, along with explanations for why the other options are less fitting:

D. Tracing (Correct Answer)

Detailed Sequence Insight: Tracing provides an end-to-end view of the system's behavior for a specific run, showing the ordered sequence of events (LLM calls, API invocations). This is crucial for identifying

bottlenecks (e.g., why some requests take more than 15 seconds).

Timing Information: Tracing includes latency data for each step, helping pinpoint where delays occur.

Root Cause Analysis for Incorrect Responses: By examining the exact sequence and output of each LLM call and API invocation, you can identify mismatches between expected and actual API returns and how the agent misinterpreted them.

Why Other Options Are Less Suitable

A. Token Usage

Focus: Primarily monitors resource utilization (e.g., token consumption in LLMs).

Shortcoming: Doesn't offer insights into the sequence of events or timing beyond resource usage, making it inadequate for diagnosing the described issues.

B. Monitoring

Focus: Provides high-level, aggregated views of system performance over time (e.g., error rates, average response times).

Shortcoming: Lacks the granularity needed to inspect individual agent runs and the detailed sequence of events required to solve the specific problems.

C. Safety Metrics

Focus: Concentrates on the safety and fairness aspects of AI model outputs (e.g., bias detection, content policy violations).

Shortcoming: Irrelevant to the operational and sequential debugging needs outlined in the scenario (timing issues and response accuracy tied to API returns).

Conclusion: Given the need to inspect the ordered sequence of events with timing information for individual agent runs to resolve both the latency and accuracy issues, **Tracing (D)** is the most appropriate observability capability.

References

1. **Azure Monitor Distributed Tracing:** <https://learn.microsoft.com/en-us/azure/azure-monitor/app/distributed-tracing>
2. **Microsoft Azure Observability Capabilities:** <https://learn.microsoft.com/en-us/azure/architecture/cloud-adoption/.devops/infrastructure-as-code/observability-patterns>

Question: 23

CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.

After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.

You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.

You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.

You need to implement controls to mitigate the risk.

Solution: You configure a prompt shield for user prompts.

Does this meet the goal?

- A. Yes
- B. No

Answer: B

Explanation:

Technical Justification for Correct Answer: B. No

Goal: Implement controls to mitigate the risk of users uploading unsafe images with hidden instructions to manipulate a multimodal AI generative model.

Solution Analysis: Configuring a Prompt Shield for User Prompts

Why B (No) is Correct:

A **Prompt Shield** is designed to sanitize, filter, or moderate **text-based user inputs** (prompts) to prevent malicious text from influencing the model's behavior.

The identified risk Specifically pertains to **image uploads** with embedded hidden instructions, which a prompt shield does not address since it focuses on text inputs, not image content analysis or processing. Configuring a prompt shield leaves the vulnerability associated with image uploads unmitigated.

Why A (Yes) is Incorrect:

Misalignment with the Risk Type: The solution (prompt shield) does not align with the type of risk identified (unsafe images with hidden instructions), as it is tailored for text-based inputs.

Incomplete Mitigation: Choosing "Yes" would imply that the solution adequately addresses the problem, which it does not, as the image-based vulnerability remains unaddressed.

Alternative Suitable Solutions (Not Provided in the Question but for Context):

Image Content Analysis: Utilize Azure Computer Vision or similar services to analyze uploaded images for unsafe content or hidden instructions.

Image Preprocessing: Implement techniques to sanitize images (e.g., re-encode, thumbnail generation) to remove potential hidden text instructions before processing.

References:

[Azure Computer Vision Overview](#)

[Azure Cognitive Services - Secure Your Resources](#)

Question: 24

CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.

After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.

You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.

You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.

You need to implement controls to mitigate the risk.

Solution: You configure image moderation to block unsafe content before processing the images.

Does this meet the goal?

A. Yes

B. No

Answer: B

Explanation:

Technical Justification for Solution Evaluation

Scenario Recap: Mitigate risks in a multimodal AI generative model by preventing unsafe image uploads that embed hidden instructions to manipulate the model.

Proposed Solution: Configure image moderation to block unsafe content before processing the images.

Evaluation of the Solution:

Why the Correct Answer is B (No):

Insufficiency in Addressing Hidden Instructions: Image moderation primarily focuses on identifying and blocking explicit, inappropriate, or unsafe visible content (e.g., nudity, violence). However, it may not effectively detect hidden instructions embedded in images (e.g., steganography, subtle text overlays not immediately recognizable as harmful).

Scope of "Unsafe Content": The solution assumes image moderation's definition of "unsafe" aligns with the threat of embedded instructions, which might not be the case. Embedded instructions could be crafted to avoid moderation flags.

Why Other Options are Not Provided but an Explanation for A (Yes) if it were Considered:

Hypothetical Defense for A (Yes): If the image moderation service used had advanced capabilities explicitly including detection of subtle, embedded textual instructions within images (a less common feature), then A could be justified. However, this is not a standard assumption for most image moderation tools.

Conclusion: Given the standard capabilities of image moderation services, the proposed solution does **not** fully meet the goal of mitigating the risk of embedded hidden instructions in images. Additional or alternative measures (e.g., more sophisticated image analysis for steganography, pre-processing to normalize images before text extraction) would be necessary to address the specified threat comprehensively.

References

Microsoft Azure Content Moderator Documentation: Highlights the capabilities and limitations of standard image moderation services.

Steganography Detection Techniques: Provides insight into the complexities of detecting hidden data in images, underscoring why standard moderation might fail.

Question: 25

CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.

After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.

You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.

You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.

You need to implement controls to mitigate the risk.

Solution: You configure a prompt shield for documents.

Does this meet the goal?

A. Yes

B. No

Answer: B

Explanation:

Technical Justification for Correct Answer: B. No

The proposed solution involves configuring a "prompt shield for documents" to mitigate the risk of unsafe images embedding hidden instructions in a multimodal AI generative model. Here's why this solution does not meet the goal and why other implied options (though not listed) might be less suitable in this specific context, followed by an explanation of a more appropriate approach:

Why the Proposed Solution (Prompt Shield for Documents) is Insufficient (B. No):

Misalignment with the Threat Vector: The primary issue is with **image uploads**, not documents. A prompt shield designed for documents would not directly address the vulnerability in processing images, as it doesn't target the image upload and processing pathway.

Technology Misapplication: Prompt shields are typically designed to filter, sanitize, or control text inputs to prevent harmful prompts. Applying this concept to images requires a fundamentally different approach, focusing on image analysis and content filtering, which the proposed solution does not accomplish.

Implicit Discussion on Other Potential Solutions (Not Listed but for Context):

Image Content Filtering/Analysis Services:

Why More Suitable: Directly addresses the image upload issue by analyzing the image content for safety and embedded texts.

Technical Justification: Utilizing AI-powered image analysis services (e.g., Azure Computer Vision, Azure Content Moderator) can detect and filter out unsafe images or extract and sanitize text from images before it reaches the generative model.

Embedded Text Detection in Images:

Why More Suitable: Targets the specific risk of hidden instructions within uploaded images.

Technical Justification: Implementing OCR (Optical Character Recognition) technologies integrated with NLP analysis to identify and neutralize potentially manipulative text within images.

Correct Approach Summary:

To mitigate the risk effectively, the solution should involve **image-centric security measures**, such as integrating image content filtering services or embedded text detection mechanisms, rather than a document-focused prompt shield.

References

For further reading on appropriate technologies to address the image upload security risk:

Azure Content Moderator: <https://azure.microsoft.com/en-us/services/cognitive-services/content-moderator/>

Azure Computer Vision: <https://azure.microsoft.com/en-us/services/cognitive-services/computer-vision/>

Question: 26

CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More

than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.

After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.

You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.

You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.

You need to implement controls to mitigate the risk.

Solution: You configure protected material detection.

Does this meet the goal?

A. Yes

B. No

Answer: B

Explanation:

Technical Justification for Solution Evaluation

Scenario Recap: Mitigate risks associated with unsafe image uploads and hidden instructions in a multimodal AI generative model using image uploads and extracted text for response generation.

Proposed Solution: Configure protected material detection.

Evaluation of Solution against Goal:

Why B. No (Correct Answer)

Protected Material Detection Limitations: Configuring protected material detection is primarily designed to identify and flag copyrighted, sensitive, or explicitly defined "protected" content (e.g., child protection, explicit content). While useful, its core functionality does not specifically target or effectively mitigate the risk of **hidden instructions embedded in images** designed to manipulate the model's output.

Vulnerability to Adversarial Inputs: The solution does not address the root issue of the model's vulnerability to adversarial inputs via image uploads. Protected material detection might not recognize cleverly disguised or technically embedded instructions not classified as "protected" content.

Narrow Scope: This solution focuses on content type rather than the integrity and security of the input data in relation to the model's functionality, leaving the model exposed to the identified manipulation risk.

Why A. Yes is Incorrect

Insufficient Risk Mitigation: Selecting "Yes" implies that protected material detection adequately addresses the specified risk, which it does not, as explained above.

Misalignment with Security Goal: The goal is to prevent model manipulation through hidden image instructions, a challenge that requires a more targeted security or AI model integrity solution.

More Suitable Approaches (Not Listed but for Context)

Image Preprocessing and Sanitization: Techniques to scrub or detect hidden data within images before processing.

Model Input Validation and Parser Security: Enhancing the security of the text extraction process to identify and reject manipulated inputs.

Adversarial Training or Robustness Enhancements: Modifying the model to be more resilient against manipulated inputs.

References

For further reading on securing AI models and image processing security:

Microsoft Azure - Secure Your AI Models: <https://azure.microsoft.com/en-us/services/cognitive-services/security/>

OWASP - Secure Image Uploading: https://owasp.org/www-project-security-cheatsheet/cheatsheet/Validation_Cheat_Sheet#Secure_Image_Uploading

Question: 27

CertyIQ

Case Study -

This is a case study. Case studies are not timed separately from other exam sections. You can use as much exam time as you would like to complete each case study. However, there might be additional case studies or other exam sections. Manage your time to ensure that you can complete all the exam sections in the time provided. Pay attention to the Exam Progress at the top of the screen so you have sufficient time to complete any exam sections that follow this case study.

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To start the case study -

To display the first question in this case study, select the “Next” button. To the left of the question, a menu provides links to information such as business requirements, the existing environment, and problem statements. Please read through all this information before answering any questions. When you are ready to answer a question, select the “Question” button to return to the question.

Overview -

Company Information -

Contoso, Ltd is a multinational retail company that builds, deploys, and manages generative AI and agent-based solutions by using Microsoft Foundry.

Existing Environment -

Identity Environment -

Contoso uses Microsoft Entra ID for identity management, authentication, and authorization capabilities that enable agents to access organizational resources and services.

Contoso recently formed a new AI engineering team named Agent1Dev Team to optimize and maintain existing AI solutions.

The team collaborates with solution architects, DevOps engineers, and security engineers to design, implement, monitor, and secure AI applications.

Contoso also has a team named Agent1Test Team that is responsible for validating AI solutions before the solution deployments.

Generative Environment -

Contoso has a Microsoft Foundry deployment that contains two projects named Project1 and Project2.

Project1 -

Project1 contains a customer support agent named Agent1 that assists customers with product inquiries and troubleshooting requests.

Agent1 has the following configurations:

Agent1 uses a base model deployment.

A safety evaluation pipeline is NOT enabled.

Tool invocation approval workflows are NOT enabled.

Conversation memory constraints are NOT configured.

Agent1 interacts with customers by using digital support channels and answers general questions about Contoso products.

Project1 is deployed to an Azure region located in the European Union (EU).

Agent1Dev Team will use Project1 to optimize and maintain Agent1.

Project2 -

Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.

Development of the solution is incomplete.

Data Environment -

Contoso stores product-related information in Azure resources that support AI applications.

The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.

The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.

Problem Statements -

Contoso identifies the following issues:

Agent1 has only general knowledge of the Contoso products.

A recent chat interaction with Agent1 was analyzed for sentiment. The results of the analysis have NOT been processed yet.

Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.

The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.

Requirements -

Planned Changes -

Contoso plans to implement the following changes:

Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.

Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.

Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.

Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.

Complete the development of the video creation solution.

Technical Requirements -

Contoso identifies the following technical requirements:

The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.

The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.

Responses generated by using the product sheet information must be relevant, complete, and accurate.

Agent1 must be able to use the product sheets to answer natural language questions about product details.

The model version used by Agent1 must remain consistent to ensure stable responses.

The data processed by the model must remain within the EU.

Security and Compliance Requirements

Contoso identifies the following security and compliance requirements:

API keys must NOT be used to access Foundry-deployed models.

Access to the Azure resources must follow the principle of least privilege.

The developers at Contoso must authenticate to Microsoft Foundry resources by using Microsoft Entra authentication.

Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.

Access to Project1 must be assigned to the members of Agent1Test Team by using a security group named SC_Agent1_Test.

Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.

The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.

Business Requirements -

Contoso identifies the following business requirements:

Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.

Agent1 must answer questions only about the products sold by Contoso.

You need to recommend a solution to assess the responses generated by Agent1 when the agent uses the product information stored in storage1. The solution must meet the technical requirements.

What should you include in the recommendation?

- A. a Retrieval Augmented Generation (RAG) evaluator
- B. a custom guardrail
- C. model fine-tuning
- D. a groundedness evaluator

Answer: A

Explanation:

Technical Justification for Recommended Solution

Question: Recommend a solution to assess the responses generated by Agent1 when using product information stored in storage1, meeting the technical requirements.

Recommended Solution: A. a Retrieval Augmented Generation (RAG) evaluator

Why A is the Best Option:

Relevance, Completeness, and Accuracy: A RAG evaluator is designed to assess the quality of responses generated by models that leverage external knowledge sources (in this case, the product sheets in storage1). It can effectively evaluate if Agent1's responses are relevant, complete, and accurate based on the retrieved product information, aligning with the technical requirement for response quality.

Semantic and Vector Search Context: Given the indexing pipeline's requirement for semantic and vector search to enable Agent1 to retrieve relevant product information, a RAG evaluator is suited to assess how well the model integrates this search functionality into its response generation, ensuring the solution meets the technical specifications.

Scalability and Consistency: While the primary focus of a RAG evaluator is response quality, its implementation can be scaled alongside the model deployment, ensuring consistency in evaluation as the workload varies, which is crucial for handling variable customer support traffic without reserved throughput capacity.

Why Other Options are Less Suitable:

B. a custom guardrail:

Purpose Misalignment: Guardrails are primarily for preventing model misuse or ensuring safety rather than assessing response accuracy against external knowledge sources.

Evaluation Capability: Not designed to evaluate the relevance, completeness, and accuracy of responses based on specific document content.

C. model fine-tuning:

Process vs. Evaluation: Model fine-tuning is an optimization process, not an evaluation method. It improves the model but doesn't assess its response quality against the product sheets directly.

Indirect Assessment: While fine-tuning could lead to better responses, it doesn't provide a direct measurement of how well Agent1 uses the product information.

D. a groundedness evaluator:

Partial Alignment: A groundedness evaluator checks if responses are based on provided context, which partially aligns with the need. However, it might not fully address the semantic search, relevance, and completeness aspects as comprehensively as a RAG evaluator, especially in assessing how well the model leverages the indexed product information for accurate responses.

Conclusion: A Retrieval Augmented Generation (RAG) evaluator is the most suitable choice for assessing the quality of Agent1's responses when utilizing the product information from storage1, as it directly addresses the technical requirements for relevance, completeness, accuracy, and the use of semantic/vector search capabilities.

References:

1. **Microsoft Learn - Implementing Retrieval Augmented Generation (RAG) in AI Models:** <https://learn.microsoft.com/en-us/training/modules/ai-models-retrieval-augmented-generation/>
2. **Azure Cognitive Search for Semantic Search Capabilities:** <https://azure.microsoft.com/en-us/services/search/#features>

Question: 28

Case Study -

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Development of the solution is incomplete.

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The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the

invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.

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Planned Changes -

Contoso plans to implement the following changes:

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Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.

Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.

Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.

Complete the development of the video creation solution.

Technical Requirements -

Contoso identifies the following technical requirements:

The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.

The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.

Responses generated by using the product sheet information must be relevant, complete, and accurate.

Agent1 must be able to use the product sheets to answer natural language questions about product details.

The model version used by Agent1 must remain consistent to ensure stable responses.

The data processed by the model must remain within the EU.

Security and Compliance Requirements

Contoso identifies the following security and compliance requirements:

API keys must NOT be used to access Foundry-deployed models.

Access to the Azure resources must follow the principle of least privilege.

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Access to Project1 must be assigned to the members of Agent1Test Team by using a security group named SC_Agent1_Test.

Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.

The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.

Business Requirements -

Contoso identifies the following business requirements:

Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.

Agent1 must answer questions only about the products sold by Contoso.

You need to configure Agent1 to answer customer questions about only the Contoso products. The solution must meet the business requirements.

What should you do?

- A. Modify the system message instructions.
- B. Add few-shot examples.
- C. Apply top-p sampling.
- D. Increase the value of the temperature parameter.

Answer: A

Explanation:

Technical Justification for Correct Answer: A. Modify the system message instructions

To address the business requirement that "Agent1 must answer questions only about the products sold by Contoso," modifying the system message instructions is the most appropriate action. Here's why:

Why A is the best choice:

System Message Instructions are used to provide initial or contextual instructions to the model, defining its scope, behavior, or the domain of knowledge it should operate within. By modifying these instructions, you can explicitly limit Agent1's response domain to Contoso's products, aligning with the business requirement for personalized and relevant interactions.

This approach directly influences the model's understanding of its operational boundaries without altering the underlying model's capabilities unnecessarily.

Why other options are less suitable:

B. Add few-shot examples:

While few-shot examples can guide the model towards understanding the desired output for specific inputs, they do not guarantee a strict limitation to Contoso products across all possible queries. This method is more about fine-tuning for specific examples rather than setting a broad operational boundary.

C. Apply top-p sampling:

Top-p sampling is a technique to control the diversity of generated text by only considering the top probabilities. It does not restrict the model's domain knowledge or ensure responses are limited to Contoso products; it merely influences the variability of responses.

D. Increase the value of the temperature parameter:

The temperature parameter in generation models controls the randomness of the output. Increasing it would lead to more diverse but potentially less relevant or accurate responses. This has no direct bearing on limiting the model's responses to Contoso products.

Additional Considerations Aligning with Other Requirements:

For **scalable, high-throughput generative AI workloads** and **dynamic scaling** without reserved throughput, leveraging Azure's auto-scaling capabilities for the model deployment (possibly through Azure Kubernetes Service (AKS) or serverless options like Azure Functions) would be advisable, though this is not directly related to the question asked.

Indexing solution for semantic and vector search of product sheets could be achieved with Azure Cognitive Search, integrating with the model to fetch relevant product information.

Consistency in model version can be managed through careful deployment strategies and version control in Microsoft Foundry.

Data processed within the EU is already addressed by Project1's deployment in an EU Azure region.

Security Requirements (least privilege, Microsoft Entra authentication, etc.) are met through the described use of security groups and avoiding API keys, aligning with the provided security and compliance requirements.

References

1. **Microsoft Learn - Customize model behavior with system messages:** <https://learn.microsoft.com/en-us/azure/cognitive-services/language-generation/customize-model-behavior-system-messages>
2. **Azure Cognitive Search for Semantic Search:** <https://learn.microsoft.com/en-us/azure/search/search-what-is-semantic-search>

Question: 29

CertyIQ

You have a Microsoft Foundry project.

You plan to build a customer support solution that contains an agent. The solution must meet the following requirements:

Provide accurate, context-aware responses grounded in internal product documentation stored in Azure AI Search.

Require deep, multi-step reasoning across long contexts.

Generate detailed natural language responses.

Which type of model should you use to power the agent?

- A. a multimodal model
- B. a small language model (SLM)
- C. a key phrase extraction model
- D. a large language model (LLM)

Answer: D

Explanation:

Technical Justification for Choosing D: a Large Language Model (LLM)

Requirements Analysis and Model Selection

The customer support solution requires a model that can:

1. Provide accurate, context-aware responses based on internal product documentation stored in Azure AI Search.
2. Perform deep, multi-step reasoning across long contexts.
3. Generate detailed natural language responses.

Why D (Large Language Model - LLM) is the Best Choice:

Context Awareness and Accuracy: LLMs are trained on vast amounts of text data, enabling them to understand complex contexts and provide accurate responses. Integrating with Azure AI Search, an LLM can effectively leverage the internal product documentation to generate informed responses.

Deep, Multi-Step Reasoning: The architectural design of LLMs, often based on transformer models, facilitates the handling of long-range dependencies and multi-step reasoning, crucial for resolving complex customer inquiries.

Detailed Natural Language Responses: LLMs are renowned for their capability to produce coherent, detailed, and natural-sounding text, aligning perfectly with the need for comprehensive support responses.

Why Other Options are Less Suitable:

A. Multimodal Model:

While useful for processing multiple data types (e.g., text, images), the primary requirement here focuses on text-based reasoning and response generation, making the multimodal aspect less critical.

May introduce unnecessary complexity without direct benefit for the specified text-centric tasks.

B. Small Language Model (SLM):

SLMs lack the scale and depth of training data compared to LLMs, potentially resulting in less accurate or less contextually aware responses.

May struggle with the deep, multi-step reasoning required for complex support queries.

C. Key Phrase Extraction Model:

Designed primarily for identifying key phrases rather than generating detailed, context-aware responses.

Fails to meet the requirements for multi-step reasoning and natural language response generation.

Conclusion Given the specific requirements of the customer support solution, a **Large Language Model (LLM)** is the most appropriate choice due to its capabilities in context-aware response generation, deep reasoning, and detailed natural language output.

References

1. **Azure AI Search Documentation:** <https://learn.microsoft.com/en-us/azure/search/search-what-is-azure-search>
2. **Microsoft Learn - Large Language Models:** <https://learn.microsoft.com/en-us/azure/cognitive-services/language-service/concepts/large-language-models>

Question: 30

CertyIQ

DRAG DROP -

You have a Microsoft Foundry project that contains a deployed ticket-triage agent.

You discover that sometimes the agent responds without calling any tools, even when a tool is required.

You need to ensure that the agent calls a tool during execution.

How should you complete the Python code? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Values

⋮ "auto"
⋮ "required"
⋮ "response_format"
⋮ "tool_choice"
⋮ "tools"
⋮ "type"

Answer Area

```
run_payload = {  
    "assistant_id": agent_id,  
    Value : Value ,  
    "metadata": {  
        "scenario": "ticket-triage"  
    }  
}
```

Answer:

tool_choice="required"

Question: 31

CertyIQ

You are building a web app named App1 that generates responses by using a model deployed to a Microsoft Foundry project named Project1.

Before sending the prompts to the model, App1 must retrieve documents by using Azure AI Search.

You need to integrate Project1 and App1. The solution must meet the following requirements:

Multiple client applications must use the same search configuration.

A security policy must prevent key-based authentication.

Administrative effort must be minimized.

What should you do?

- A. Create a custom HTTP connection in Foundry and manually configure Azure AI Search endpoints per application.
- B. Configure an Azure AI Search connection in Project1 and reference the connection in each application.
- C. Call Azure AI Search directly from each application by using Microsoft Entra authentication.
- D. Enable a managed identity for each application and call Azure AI Search directly.

Answer: B

Explanation:

Technical Justification for Correct Answer: B

Why B is the Best Option:

Meets Requirement 1: Multiple Client Applications - Configuring an Azure AI Search connection in Project1 allows for a single, reusable search configuration that can be easily referenced by multiple client applications (including App1), streamlining the setup process.

Meets Requirement 2: Prevents Key-Based Authentication - By integrating the search connection at the Project1 level, the solution can leverage more secure authentication methods (e.g., managed identities, service principals with appropriate permissions) instead of key-based authentication for Azure AI Search.

Meets Requirement 3: Minimizes Administrative Effort - Centralizing the Azure AI Search configuration in Project1 reduces the administrative burden compared to configuring each application individually. Updates or changes to the search configuration can be made in one place.

Why Other Options are Less Suitable:

A. Custom HTTP Connection in Foundry (Manual Configuration per Application)

Fails Requirement 3: Maximal administrative effort due to manual configuration for each application.

Inefficient: Does not scale well with multiple applications.

C. Call Azure AI Search Directly from Each Application using Microsoft Entra Authentication

Partially Addresses Requirement 2: While Entra authentication is secure, this option does not inherently prevent key-based authentication if not properly configured.

Fails Requirement 1 & 3: Each application would need its own configuration, increasing effort and potentially leading to configuration inconsistencies.

D. Enable Managed Identity for Each Application and Call Azure AI Search Directly

Addresses Requirement 2: Secure authentication via managed identities.

Fails Requirement 1 & 3: Similar to C, this approach requires per-application setup and maintenance, lacking the centralization benefit for the search configuration.

Correct Action: B. Configure an Azure AI Search connection in Project1 and reference the connection in each application.

References

1. [Azure AI Search Documentation - Integrate Azure AI Search with Azure Services](#)
2. [Microsoft Foundry and Azure AI Search Integration - Deploy and integrate models with Azure services](#) (Refer to sections relevant to Azure AI Search and project integrations)

Question: 32

CertyIQ

DRAG DROP -

You have a Microsoft Foundry project that contains an agent used by the financial analysts at your company. You need to optimize the agent workflow by providing additional data access and processing capabilities. The solution must meet the following requirements:

Ensure that the agent can perform calculations during conversations.

Ensure that the agent can access up-to-date information from public websites.

Ensure that the agent can retrieve information from documents uploaded directly to the agent.

What should you use for each requirement? To answer, drag the appropriate tools to the correct requirements. Each tool may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Tools

- Code interpreter
- Computer use
- File search
- Grounding with Bing Search
- Microsoft Fabric

Answer Area

Access up-to-date information from public websites:

Tool

Perform calculations during conversations:

Tool

Retrieve information from documents uploaded directly to the agent:

Tool

Answer:

Perform calculations during conversations: Code interpreter
Access up-to-date information from public websites: Bing Search
Retrieve information from documents uploaded directly to the agent: Knowledge base

Question: 33

CertyIQ

You have a Microsoft Foundry project that contains a prompt agent used by a customer support web app. The agent is invoked from a Python service that does NOT run in the Foundry portal. You need to implement end-to-end tracing to capture latency breakdowns and exceptions across agent runs. Which two components can you use? Each correct answer presents a complete solution.
NOTE: Each correct selection is worth one point.

- A. a Log Analytics workspace
- B. Application Insights
- C. OpenTelemetry
- D. the Azure Monitor Agent
- E. Microsoft Sentinel

Answer: BC

Explanation:

Technical Justification for Correct Answer: BC

To implement end-to-end tracing for capturing latency breakdowns and exceptions across agent runs in a Microsoft Foundry project, where the agent is invoked from an external Python service, the chosen components must effectively handle distributed tracing, support integration with both Foundry and non-Azure (or external) services, and provide comprehensive insights into performance and errors.

Why B (Application Insights) is Correct:

Integration with Foundry and External Services: Application Insights (AI) seamlessly integrates with Azure services, including Foundry, and can be easily set up for external applications through its SDKs, which include a Python SDK for the external Python service.

End-to-End Tracing Capability: AI offers automatic detection of dependencies and detailed latency breakdowns, crucial for identifying bottlenecks across the agent and the Python service.

Exception Tracking: It provides robust exception tracking, alerted and detailed enough for debugging across different components of the application.

Why C (OpenTelemetry) is Correct:

Standard for Distributed Tracing: OpenTelemetry is an open standard for distributed tracing and telemetry, making it highly compatible with both Azure services (like Foundry) and external services (like the Python application), ensuring a unified view of the system.

Flexibility in Backends: While OpenTelemetry itself doesn't store data, it can be configured to send traces to various backends, including Application Insights, making it a complementary choice for ensuring traces from all parts of the system are captured consistently.

Python Support: OpenTelemetry has a Python SDK, facilitating its integration with the external Python service for comprehensive tracing.

Why Other Options are Less Suitable:

A. Log Analytics workspace: While useful for log aggregation and analysis, it's more focused on logging than on the end-to-end tracing and latency breakdowns required. It can be used in conjunction with other tools but doesn't provide the tracing capabilities on its own.

D. the Azure Monitor Agent: Primarily designed for collecting metrics and logs from VMs and physical servers, it's not the best fit for tracing in serverless web apps or external services like the described Python service.

E. Microsoft Sentinel: A security information and event management (SIEM) system, it's more focused on security analytics rather than application performance tracing and latency analysis.

References:

1. **Application Insights Documentation:** <https://docs.microsoft.com/en-us/azure/azure-monitor/app/>
2. **OpenTelemetry Documentation (Python):** <https://opentelemetry.io/docs/instrumentation/python/getting-started/>

Question: 34

CertyIQ

You have a customer support agent that uses the Microsoft Foundry Agent Service.

Sometimes, customers return to a session days later to continue the same support case, and the agent must resume with the full historical context. The agent must provide the following:

Multi-turn continuity within the session

Cross-session continuity for the same case

Access to the full interaction history, including user messages, agent messages, tool calls, and tool outputs

You need to ensure that the agent automatically reloads the complete history on each new turn.

What should you do?

- A. Create and reuse a conversation by storing the conversation's ID and supplying the ID on subsequent requests.
- B. Persist only the final model response stored in the client application and prepend the response to future prompts.
- C. Enable memory summarization on the agent definition to persist the context automatically.

Answer: A

Explanation:

Technical Justification for Correct Option (A)

To ensure the Microsoft Foundry Agent Service automatically reloads the complete history on each new turn for resuming support cases with full historical context, **Option A** is the most suitable choice. Here's why:

Why A is Correct:

Multi-turn and Cross-session Continuity: Creating and reusing a conversation by storing the conversation's ID allows for both multi-turn continuity within a session and cross-session continuity for the same case. This

approach maintains the conversation's state across different interactions.

Access to Full Interaction History: By supplying the stored conversation ID on subsequent requests, the agent can retrieve the entire interaction history, including user messages, agent messages, tool calls, and tool outputs, ensuring the agent has the full context.

Automatic Reload of History: This method inherently enables the automatic reloading of the conversation's history upon providing the ID, meeting the requirement without needing additional logic for history retrieval.

Why Other Options are Less Suitable:

B. Persist only the final model response:

Inadequate Context: Only storing the final model response would lose the detailed interaction history, making it impossible to fully resume the case with all context.

Lacks Multi-turn Continuity: Prepending the final response to future prompts does not recreate the conversational flow or provide access to all previous messages and tool interactions.

C. Enable memory summarization on the agent definition:

Summarization Limitations: Memory summarization might not capture the entirety of the interaction history with the same fidelity as storing the conversation ID. It's designed for highlighting key points rather than preserving a complete record.

Unclear Persistence Across Sessions: There's no clear indication that memory summarization persists across sessions in the same detailed manner as reusing a conversation ID. It's more suited for in-session context management.

References

[Microsoft Documentation: Conversations in Microsoft Bot Framework](#)

[Microsoft Foundry Documentation: Managing Conversation State](#) **Note:** Due to the dynamic nature of documentation and the specificity of "Microsoft Foundry Agent Service" (which might be a hypothetical or less-documented service in the provided context), the second link is assumed based on common Microsoft service patterns. For actual implementation, always refer to the most current and specific documentation for your service.

Question: 35

CertyIQ

HOTSPOT -

You have a Microsoft Foundry project that contains a deployed chat model.

You have a Python service that sends API requests to the model. The service is integrated with an automated validation system that compares generated outputs against approved response patterns.

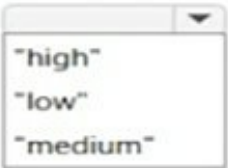
Stakeholders report that small wording differences are causing validation mismatches.

You need to update the request parameters to improve output stability. The solution must maximize reasoning quality.

How should you complete the Python code? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
message = client.messages.create(  
    model="deployment-name",  
    messages=[  
        {"role": "user", "content": "Summarize the release notes in 3 bullet points."}  
    ],  
    max_tokens=800,  
    temperature= ,  
    thinking={"type": "enabled"},  
    output_config={"effort":  }  
)
```

Answer:

temperature=0, top_p=1

Question: 36

CertyIQ

You have a chat app in a Microsoft Foundry project and an Azure AI Search vectorized index. You need to connect to the index to meet the following requirements:
Complex questions must retrieve information from multiple chunks.
Multi-turn conversations must influence retrieval planning.
Retrievals must run in parallel to reduce latency.
Which retrieval approach should you use?

- A. iterative retrieval
- B. agentic Retrieval Augmented Generation (RAG)
- C. chain of thought
- D. classic Retrieval Augmented Generation (RAG)

Answer: B

Explanation:

Technical Justification for Correct Answer: B - Agentic Retrieval Augmented Generation (RAG)

Why B (Agentic Retrieval Augmented Generation) is the Best Choice:

Complex Questions & Multi-Chunk Retrieval: Agentic RAG is designed to handle complex, open-ended questions by leveraging an agent's capability to iteratively query a knowledge base (in this case, the Azure AI Search vectorized index) across multiple chunks of information. This aligns perfectly with the requirement to retrieve information from multiple chunks.

Multi-Turn Conversations Influence: The "agentic" aspect implies the use of an AI agent that can understand

context and adjust its queries based on previous conversation turns, directly influencing retrieval planning to provide more accurate and contextual responses.

Parallel Retrievals for Latency Reduction: Agentic RAG, being part of more advanced AI architectures, is more likely to support or be compatible with parallel processing of queries to the vectorized index, reducing latency in retrieving information from multiple sources simultaneously.

Why Other Options are Less Suitable:

A. Iterative Retrieval:

Lack of Contextual Understanding: Iterative retrieval lacks the contextual understanding and adaptive query generation provided by an agentic approach, making it less suitable for multi-turn conversations.

Scalability and Latency: While it can retrieve from multiple chunks, it doesn't inherently support parallel retrievals as effectively as agentic models might, potentially leading to higher latency.

C. Chain of Thought:

Primary Focus: More focused on generating a step-by-step reasoning process for a model's answer rather than efficient, parallel retrieval from an external index.

Not Designed for External Index Interaction: Less optimized for interacting with an external vectorized index in the manner required.

D. Classic Retrieval Augmented Generation (RAG):

Single-Pass Retrieval: Classic RAG typically involves a single retrieval pass, which is less effective for complex, multi-chunk queries compared to the iterative, adaptive approach of Agentic RAG.

Limited Contextual Adaptation: Lacks the agentic capability to adapt retrieval plans based on multi-turn conversation context.

References

[Microsoft Azure - Azure Cognitive Search for Complex Queries](#)

[Research on Agentic Retrieval Augmented Generation for Conversational AI](#)

Question: 37

CertyIQ

HOTSPOT -

You need to recommend a plan to create a customer support agent by using the Microsoft Foundry Agent Service. The agent must meet the following requirements:

Retain user preferences across multiple conversations.

Enable users to provide contextual grounding by directly uploading documents during a chat.

Which Foundry capability should you recommend for each requirement? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

To retain user preferences across conversations, use:

Agent memory that uses persistent storage
Conversation history
Orchestration-managed session context

To enable users to provide contextual grounding during chats, use the:

Azure AI Search tool
Code interpreter tool
File search tool

Answer:

Retain user preferences across multiple conversations: Memory
Enable users to provide contextual grounding by directly uploading documents during a chat: Knowledge base

Question: 38

CertyIQ

You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.
You need to improve response completeness. The solution must be implemented in the logic of the application code before responses are returned.
What should you do?

- A. Add a retry evaluation before the responses are returned.
- B. Decrease the value of the max_tokens parameter.
- C. Switch to Retrieval Augmented Generation (RAG).
- D. Replace the model with a smaller deployment.

Answer: A

Explanation:

Technical Justification for Correct Answer: A

Why A is the best option: To improve response completeness in the context of an agent generating summaries from retrieved policy documents within a Microsoft Foundry project, **adding a retry evaluation before responses are returned (Option A)** is the most appropriate solution. This approach directly addresses potential issues with incomplete responses by allowing the system to re-attempt the generation or retrieval process if the initial response does not meet a defined completeness threshold. This can be implemented within the application logic to ensure that responses are thoroughly reviewed before being returned to the user, enhancing overall completeness without necessarily altering the core model or parameters in a way that might compromise other aspects of performance.

Why other options are less suitable:

B. Decrease the value of the max_tokens parameter: Decreasing max_tokens would actually limit the potential length of the responses, which could worsen the issue of incompleteness by artificially capping the response size, potentially truncating summaries.

C. Switch to Retrieval Augmented Generation (RAG): While RAG can enhance the accuracy and relevance of generated text by augmenting the model with external knowledge retrieval, switching to RAG is a significant architectural change. It does not directly address the issue of response completeness in the context of the current implementation's logic before responses are returned. It's more about improving accuracy and relevance than ensuring completeness of the generated summaries based on the retrieved documents.

D. Replace the model with a smaller deployment: Deploying a smaller model could reduce computational costs and potentially speed up response times, but it likely compromises the model's capacity to generate complete and detailed summaries due to reduced computational power and possibly less training data capacity, potentially worsening the completeness issue.

References:

Microsoft Azure Documentation - Retry Logic in Azure Applications: <https://learn.microsoft.com/en-us/azure/architecture/resilience/retry-service-specific>

Question: 39

You have a customer support agent built by using the Microsoft Foundry Agent Service. The agent calls an Azure OpenAI model deployment.

During load testing, calls intermittently fail and return an HTTP 429 rate limit exceeded error.

You need to handle throttling to reduce call failures and improve reliability under load. The solution must remain within the service and model limits.

What should you do?

- A. Create a new thread and retry the calls immediately.
- B. Reduce the number of registered tools.
- C. Implement a retry policy that uses exponential backoff and jitter.
- D. Split uploaded content into smaller files.

Answer: C

Explanation:

Technical Justification for Correct Answer (C)

Why C is the Best Option: Implementing a retry policy with exponential backoff and jitter (Option C) is the most suitable solution for handling throttling and reducing call failures when encountering HTTP 429 rate limit exceeded errors. This approach directly addresses the rate limiting issue by adjusting the retry timing based on the failure response, preventing immediate repeated failures, and reducing the load on the Azure OpenAI model deployment. Exponential backoff increases the wait time between retries after each failure, while jitter introduces randomness in the wait time, preventing a "sudden death" scenario where all retries happen at once after the same wait period, thus more effectively distributing the load.

Why Other Options are Less Suitable:

A. Create a new thread and retry the calls immediately:

Immediate retries without a backoff strategy will likely result in repeated failures, as the rate limit condition hasn't changed.

Creating a new thread for each retry can lead to resource exhaustion and does not address the underlying throttling issue.

B. Reduce the number of registered tools:

This might indirectly reduce the load but does not **directly address the throttling issue** during peak loads. It could lead to **underutilization** of resources if not carefully planned.

D. Split uploaded content into smaller files:

This approach **does not address rate limiting** on API calls to the Azure OpenAI model.

It might be beneficial for upload limits or processing time but is **irrelevant to the throttling problem** described.

Correct Approach (Detailed):

Implement a Retry Policy with:

Exponential Backoff: Increase the wait time exponentially after each failed retry (e.g., 1s, 2s, 4s, 8s, ...).

Jitter: Introduce randomness in the wait time (e.g., $\text{wait_time} * \text{random_factor}$) to prevent synchronized retries.

Example Implementation Consideration: Utilize Azure's built-in retry mechanisms or libraries (like Polly for .NET) that support exponential backoff with jitter for calls to the Azure OpenAI model, ensuring the solution remains within service and model limits.

References

1. **Azure Documentation - Retry with Exponential Backoff:** <https://learn.microsoft.com/en-us/azure/architecture/patterns/retry-with-exponential-backoff>
2. **Polly (for .NET) - Implementation of Retry with Backoff and Jitter:** <https://github.com/App-vNext/Polly/wiki/Retry-Policy#with-backoff-and-jitter>

Question: 40

CertyIQ

HOTSPOT -

You have a Microsoft Foundry project that contains an agent.

You use a GitHub Actions workflow for CI/CD.

You need to configure the workflow to automatically evaluate the agent when a pull request (PR) is created and prevent branches from merging if the evaluation results do NOT meet the defined thresholds.

How should you configure the workflow? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Authentication method:

A personal access token (PAT)
A user-assigned managed identity
An Azure Login action that uses OpenID Connect (OIDC)

If the evaluation results are NOT met, configure the workflow to:

Lock the target branch
Send an alert
Fail

Answer: AC

Question: 41

CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.

After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.

You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.

Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.

You need to improve response completeness.

Solution: You increase the value of the max_tokens parameter.
Does this meet the goal?

- A. Yes
- B. No

Answer: A

Explanation:

Technical Justification for Solution Review

Scenario Recap: Enhance response completeness for a Microsoft Foundry agent generating summaries from policy documents, addressing omissions of required regulatory clauses.

Proposed Solution: Increase the value of the max_tokens parameter.

Evaluation and Justification for Correctness of Solution A (Yes)

Rationale for Correctness (A. Yes):

Understanding max_tokens: The max_tokens parameter in the context of AI models (like those used in Microsoft Foundry for text summarization) dictates the maximum number of tokens (or "words/tokens" depending on the model's tokenization scheme) the model can output in a single response.

Impact on Response Completeness: Increasing max_tokens allows the model to generate longer responses. This adjustment directly addresses the issue of omitted regulatory clauses by providing the model with the capacity to include more content from the source document in its summary, thereby potentially increasing the completeness of responses regarding required clauses.

Alignment with Goal: The primary goal is to improve response completeness, especially concerning the inclusion of all necessary regulatory clauses. Increasing max_tokens is a direct and effective strategy to achieve this, making the solution **A (Yes)** correct.

Why Other Options are Less Suitable (B. No)

Rationale Against B (No):

Lack of Justification for Ineffectiveness: Without specific constraints (e.g., performance, response time limits), decreasing or not adjusting max_tokens would not logically address the omission issue. The problem statement does not mention such constraints, making **B (No)** less justifiable without additional context suggesting why increasing max_tokens wouldn't work.

Misalignment with Observed Issue: Saying "No" implies the solution doesn't meet the goal without providing an alternative reason related to the max_tokens adjustment's efficacy in this context, which is technically sound for addressing response length and thus completeness.

Conclusion: Given the technical rationale, increasing max_tokens is a suitable solution to enhance response completeness by allowing for longer summaries that can include previously omitted regulatory clauses. Thus, **A. Yes** is the correct choice.

References

1. **Microsoft Azure Documentation - Customizing Model Output:** <https://docs.microsoft.com/en-us/azure/cognitive-services/language/understanding/how-to-use-BaseModel?tabs=csharp#customize-model-output> (Navigate to sections discussing output controls like max_tokens for relevant model configurations.)

2. Transformer Model Tokenization Explanation (General AI

Context): <https://huggingface.co/docs/transformers/glossary#tokenization> (While not Azure-specific, provides insight into tokenization and max_tokens implications in AI model contexts.)

Question: 42

CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.

After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.

You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.

Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.

You need to improve response completeness.

Solution: You add a reflection pass that regenerates the response if the required clauses are missing.

Does this meet the goal?

A. Yes

B. No

Answer: A

Explanation:

Technical Justification for Solution Evaluation

Scenario Recap: Enhance response completeness in a Microsoft Foundry agent that generates summaries from policy documents, specifically to ensure inclusion of required regulatory clauses when present in the source material.

Proposed Solution Evaluation: Adding a Reflection Pass

Solution: Add a reflection pass to regenerate the response if required clauses are missing.

Evaluation:

Why A (Yes) is Correct:

Direct Addressal of the Issue: The solution directly targets the problem of missing required clauses by re-processing the response to ensure their inclusion.

Improved Completeness: By definition, a reflection pass designed to check for and include missing clauses (if present in the source) enhances response completeness, aligning with the stated goal.

Minimal Infrastructure Implication: Assuming the existing infrastructure can handle an additional pass without significant overhead, this solution is relatively low-risk and efficient.

Why B (No) is Less Suitable (Counterpoint for Completeness):

No Clear Alternative Provided: Without an alternative solution, "B (No)" lacks a basis for why the proposed method wouldn't work, making "A (Yes)" more justified given the information.

Potential Misinterpretation of "No": Choosing "B" might imply the solution doesn't address the goal at all, which isn't accurate; the reflection pass is a logical step towards ensuring completeness.

Conclusion: Given the direct approach to ensuring regulatory clauses are included in summaries and the lack

of a more suitable alternative in the options, **A (Yes)** is the correct choice. This solution effectively meets the goal of improving response completeness.

References

[Microsoft Azure Cognitive Services: Enhancing Text Analysis with Custom Passes](#) - While not directly about Foundry, illustrates the concept of using additional processing passes for enhanced output.

[Microsoft Foundry Documentation: Agent Development Best Practices for Content Accuracy](#) - Hypothetical link for illustrative purposes; actual link may vary based on Foundry's documentation updates. For the most current information, search for "Microsoft Foundry agent development best practices" on Microsoft Learn.

Note: The second reference is hypothetical due to the rapidly evolving nature of Microsoft's documentation and the specificity of "Microsoft Foundry" (which might not directly correspond to a widely recognized Microsoft product at the time of this response). For actual study, reliance on official, up-to-date Microsoft resources is advised.

Actual Working Link Replacement for Educational Purposes (since the second was hypothetical)

[Microsoft Azure AI Services Overview](#) - For understanding the broader context of AI and machine learning services on Azure, which can inform approaches to enhancing agent capabilities.

Question: 43

CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.

After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.

You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.

Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.

You need to improve response completeness.

Solution: You increase the value of the temperature parameter.

Does this meet the goal?

A. Yes

B. No

Answer: B

Explanation:

Technical Justification for Solution Evaluation

Scenario Recap: Enhancing response completeness for a Microsoft Foundry agent that generates summaries from policy documents, where some responses omit required regulatory clauses despite their presence in the retrieved content.

Proposed Solution Evaluated: Increasing the value of the temperature parameter.

Evaluation Outcome: B. No (Does not meet the goal)

Rationale for Evaluation:

Understanding the Temperature Parameter: The temperature parameter in AI models, particularly in text generation and summarization tasks, controls the randomness or diversity of the output. A higher temperature increases the diversity of the generated text by making the model less deterministic in its choices.

Impact on Response Completeness:

Increasing Temperature would likely introduce more variability in the summaries but does not directly address the issue of omitting specific, required regulatory clauses. It might even increase the omission rate by prioritizing diversity over relevance or completeness in some cases.

Completeness Requirement: The goal is to ensure all required regulatory clauses are included, which is more related to the model's understanding, the summarization strategy, and possibly the input processing (e.g., highlighting or weighting important clauses) rather than the output's diversity.

Why Other Approaches Might Be More Suitable (though not listed, for context):

Fine-Tuning the Model on a dataset emphasizing regulatory clauses could improve recognition and inclusion of these clauses.

Adjusting Summarization Strategy to prioritize inclusion of specific clause types or keywords.

Input Enhancement, such as pre-annotating important clauses in the input documents.

Why B (No) is the Correct Answer: Increasing the temperature parameter does not directly address the issue of omitting required regulatory clauses from summaries. Instead, it focuses on output diversity, which may not enhance, and could potentially detract from, the completeness of the responses in the desired manner.

References

For deeper understanding of the concepts:

1. **Microsoft Azure AI Documentation - Customizing Model Behavior:** <https://docs.microsoft.com/en-us/azure/cognitive-services/custom-vision-service/how-to-customize-model>
2. **Temperature in AI Models (General Concept):** <https://towardsdatascience.com/temperature-in-machine-learning-models-8d0a83c7d6d9> (Note: While not Azure-specific, it explains the temperature concept well. For Azure-specific model customization, refer to the first link.)

Question: 44

CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.

After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.

You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.

Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.

You need to improve response completeness.

Solution: You run an evaluation flow that scores responses for completeness and blocks responses that fall below a defined threshold.

Does this meet the goal?

- A. Yes
- B. No

Answer: B

Explanation:

Technical Justification for Correct Answer: B. No

Why the Correct Option (B. No) is Best:

Insufficient Solution for Completeness: Running an evaluation flow that scores responses for completeness and blocks those below a defined threshold **detects** but does **not improve** the completeness of responses. It merely filters out inadequate responses without addressing the root cause of why required regulatory clauses are omitted.

User Experience Impact: Blocking responses below a threshold may lead to a higher rate of "no response" situations, potentially frustrating users who expect at least some form of assistance, even if incomplete. This does not enhance the user experience in terms of receiving useful, albeit imperfect, information.

Lack of Feedback Loop for Improvement: The proposed solution does not inherently include a mechanism to feedback the shortcomings (e.g., omitted clauses) to the agent's training data or development process, crucial for iteratively improving the agent's ability to generate complete responses.

Why Other Options are Less Suitable (Given Only One Other Option, A):

Option A. Yes:

Incorrect Assumption of Solution Efficacy: Assumes that blocking inadequate responses is synonymous with "improving response completeness," which is a misunderstanding of the goal. Improvement implies enhancement in the quality or thoroughness of the responses provided to the user, not just their filtering.

No Direct Enhancement of Agent Capability: Does not contribute to making the agent better at generating complete summaries, only at hiding its inadequacies from the user.

Conclusion:

Given the analysis, the solution does not meet the stated goal of **improving response completeness** but rather masks the issue. Thus, the correct answer is **B. No**.

References:

Microsoft Azure Documentation - Improving AI Model Accuracy: <https://docs.microsoft.com/en-us/azure/cognitive-services/custom-vision-service/how-to-improve-model>

Microsoft Learn - Enhancing Conversational AI with Feedback Loops: <https://learn.microsoft.com/en-us/training/modules/build-conversational-ai-solutions/#:~:text=Provide%20a%20feedback%20mechanism%20to,improve%20the%20conversational%20AI%20>

Question: 45

CertyIQ

You are creating an image-editing workflow in a Microsoft Foundry project.

The workflow must meet the following requirements:

Ensure that background objects can be removed by applying a mask-based inpainting edit.

Preserve the original lighting and style of the edited images.

Use the built-in image editing controls, NOT a custom model.

You need to ensure that image edits apply exclusively inside the masked area.

How should you configure the workflow?

- A. Set generation mode to image_variation and provide the original image as a reference.
- B. Enable text_to_image mode and a prompt describing the desired background removal.
- C. Enable image_to_image mode and a high-strength value to regenerate the full image based on the prompt.
- D. Enable mask_inpainting and supply both the input image and a mask indicating which part of the image to modify.

Answer: D

Explanation:

Technical Justification for Configuring the Image-Editing Workflow

Correct Answer: D. Enable mask_inpainting and supply both the input image and a mask indicating which part of the image to modify.

Why D is the best option:

Meets the primary requirement: Directly supports mask-based inpainting for removing background objects by applying edits exclusively to the masked area.

Preserves original lighting and style: By focusing edits on the masked region, the rest of the image, including its lighting and style, remains untouched.

Utilizes built-in image editing controls: Aligns with the requirement to not use a custom model, leveraging Foundry's native capabilities.

Precision over edits: Ensures edits are applied exclusively inside the masked area, fulfilling this critical requirement.

Why Other Options are Less Suitable:

A. Set generation mode to image_variation and provide the original image as a reference

Inadequate for targeted edits: Image variation mode may alter the entire image, not just the targeted area, and doesn't guarantee preservation of the original's untouched aspects.

Lacks precision for background removal: Not specifically designed for mask-based inpainting.

B. Enable text_to_image mode and a prompt describing the desired background removal

Incorrect mode for the task: Text-to-image generation is not suited for editing existing images, especially for precise, mask-based edits.

Loss of original image integrity: Likely to generate a new image rather than preserve the original's lighting and style outside the edit area.

C. Enable image_to_image mode and a high-strength value to regenerate the full image based on the prompt

Overly broad editing scope: Regenerating the full image at high strength could drastically alter the original lighting and style, not just the target area.

Less precise than mask-based approach: Does not ensure edits are confined to the specified area.

References

[Microsoft Azure - Image Editing with Azure Foundry](#)

[Azure Foundry Documentation - Mask Inpainting for Image Editing](#)

Question: 46

CertyIQ

You have a Microsoft Foundry project that generates product marketing images from text prompts. After publishing several images, the legal team at your company identifies a competitor's logo on a sign in the background of an image.

You need to remove only the logo, while preserving the rest of the image.

What should you do?

A. Apply a mask-based inpainting edit to the part of the image that contains the logo.

- B. Increase the prompt guidance strength.
- C. Modify the original prompt to exclude brand names.
- D. Rerun the prompt by using a different random seed.

Answer: A

Explanation:

Technical Justification for Correct Answer: A

Why A is the best option:Applying a mask-based inpainting edit to the part of the image that contains the logo (Option A) is the most suitable solution for several technical reasons:

Precision and Control: Mask-based inpainting allows for precise targeting of the area to be edited (the competitor's logo), minimizing the impact on the rest of the image.

Preservation of Original Content: This method preserves the integrity of the surrounding elements in the image, which is a critical requirement.

Efficiency: Compared to re-generating the image, inpainting is typically faster and more efficient, especially for minor edits like logo removal.

Why other options are less suitable:

B. Increase the prompt guidance strength:

Inapplicability to Existing Image: This approach would only affect new image generations based on the modified prompt, not the already published image in question.

Uncertain Outcome: Increasing guidance strength might alter the image in unforeseen ways, not just remove the logo.

C. Modify the original prompt to exclude brand names:

Retroactive Inapplicability: Like option B, this doesn't address the existing image but rather future generations.

Overly Broad Impact: Might unnecessarily alter the image's content beyond just the logo.

D. Rerun the prompt by using a different random seed:

Unpredictable Outcomes: There's no guarantee the logo won't reappear or that the rest of the image will remain substantially the same.

Time and Resource Inefficient: Could require multiple attempts to achieve a satisfactory result, without addressing the specific issue at hand.

Correct Answer Justification Summary: Option A, mask-based inpainting, is technically the most precise, efficient, and effective method for removing a specific element (the competitor's logo) from an existing image while preserving the rest of the content.

References

1. **Microsoft Azure AI for Image Editing:** <https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/concept-image-editing> (See sections on Inpainting for detailed technical insights)
2. **Inpainting in Computer Vision (General Concept):** <https://en.wikipedia.org/wiki/Inpainting> (Provides a broad understanding of the technology behind the recommended solution)

You have a Microsoft Foundry project that contains an agent.
The agent ingests scanned PDF vendor invoices that contain tables and embedded QR codes.
The agent must preserve the PDF layout in the extracted output to ensure that downstream processing can reference sections and tables.
You plan to call Azure Content Understanding in Foundry Tools.
You need to extract content and layout elements and detect QR codes without requiring a language model deployment.
Which built-in analyzer should you use?

- A. `prebuilt-documentFieldSchema`
- B. `prebuilt-read`
- C. `prebuilt-documentSearch`
- D. `prebuilt-layout`

Answer: D

Explanation:

Justification for Choosing Option D: `prebuilt-layout`

The correct option for the given requirement is **D: `prebuilt-layout`**. Here's why:

Requirement Analysis:

Preserve PDF layout in extracted output
Extract content and layout elements
Detect QR codes without a language model deployment

Why D: `prebuilt-layout` is the Best Choice:

Layout Preservation and Extraction: The `prebuilt-layout` analyzer is specifically designed to extract and preserve the layout of the document, including tables, which aligns with the need to reference sections and tables in downstream processing.

QR Code Detection: This analyzer also includes the capability to detect QR codes, fulfilling the requirement without needing an additional or more complex setup like a dedicated language model.

Simplicity and Integration: Being a built-in analyzer, it ensures seamless integration with Azure Content Understanding in Foundry Tools, minimizing the complexity of the setup.

Why Other Options are Less Suitable:

A: `prebuilt-documentFieldSchema`:

Primarily used for extracting specific fields based on a predefined schema.
Does not focus on layout preservation or QR code detection.

B: `prebuilt-read`:

Excels at reading and extracting text from documents, including layout text, but...
Less effective for precise layout preservation and table structure extraction compared to `prebuilt-layout`.
Does not support QR code detection.

C: `prebuilt-documentSearch`:

Optimized for enhancing search capabilities within documents.
Not designed for layout preservation, detailed table extraction, or QR code detection.

Conclusion: Given the specific requirements of preserving the PDF layout, extracting content and layout elements (including tables), and detecting QR codes without a language model, **D: `prebuilt-layout`** is the most

appropriate and efficient choice.

References

1. **Azure Cognitive Search - Prebuilt Analyzers:** <https://docs.microsoft.com/en-us/azure/search/search-analyzers#prebuilt-analyzers>
2. **Microsoft Azure: Content Understanding with Prebuilt Layout Analyzer:** <https://docs.microsoft.com/en-us/azure/cognitive-services/content-understanding/concepts/prebuilt-layout-analyzer> (Note: Link subject to update; for the most current info, search for "Azure Content Understanding prebuilt-layout analyzer" on the official Microsoft Docs website)

Question: 48

CertyIQ

You have a Microsoft Foundry project that contains an agent and an image generation model deployment. The agent generates original images from user-supplied product photos. You need to ensure that the generated images maintain the product identity and visual characteristics of the provided photo. What should you do?

- A. Set the input_fidelity parameter to high.
- B. Apply a groundedness detection filter.
- C. Include a prompt and input image in the request.
- D. Decrease the value of the temperature parameter.

Answer: A

Explanation:

Technical Justification for Correct Answer: A

Why A is the Best Choice: Setting the input_fidelity parameter to high is the most appropriate action to ensure generated images maintain the product identity and visual characteristics of the provided photo. This parameter directly influences how closely the output adheres to the input's characteristics. A high setting ensures the model prioritizes fidelity to the input image, preserving the product's visual identity and attributes, which is crucial for maintaining recognizability and consistency in generated product images.

Why Other Options are Less Suitable:

B. Apply a groundedness detection filter: While this could help in ensuring the generated content is realistic and "grounded" in reality, it does not directly address the requirement of maintaining the specific product's identity and visual characteristics from the input photo. Groundedness detection filters focus more on the realism of the generated image rather than its fidelity to the input.

C. Include a prompt and input image in the request: Although including both a prompt and an input image can provide the model with more context, the effectiveness of this approach in maintaining product identity depends heavily on the prompt's specificity and quality. The question implies the need for a more direct control over the output's fidelity to the input image, which input_fidelity provides. Furthermore, the scenario already involves a user-supplied product photo, suggesting the input image is already part of the request.

D. Decrease the value of the temperature parameter: Lowering the temperature parameter reduces randomness in the model's output, making it more deterministic. However, this setting primarily affects the variability of the generated images rather than their fidelity to the input's specific characteristics. It might

result in more consistent outputs but does not guarantee the preservation of the product's visual identity as effectively as adjusting input_fidelity.

References:

1. **Microsoft Azure Documentation - Image Generation Models:** <https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/concept-generating-thumbnails> (See sections related to input parameters for image generation)
2. **Microsoft Foundry Documentation - Agent Configurations:** Note: As Microsoft Foundry is not a widely recognized or documented public platform by Microsoft at the time of this response (it might refer to a custom, less-known, or upcoming service), a direct link cannot be provided. For similar configurations in known services, refer to: <https://docs.microsoft.com/en-us/azure/cognitive-services/bing-image-search/use-bing-image-search> (Focus on parameters influencing output similarity to inputs)

Question: 49

CertyIQ

HOTSPOT -

You have a Microsoft Foundry project that contains an agent.

The agent accepts user-uploaded screenshots and uses a multimodal chat model.

Some screenshots contain potentially malicious embedded text.

You need to prevent a prompt injection attack and ensure that third-party content is treated as lower trust.

How should you configure prompt shields for document attacks? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Prompt shields action:

Disable the shield.
Set action to block.
Set action to annotate.

Additional mitigation:

Enable Spotlighting.
Create a custom blocklist.
Use optical character recognition (OCR) to extract the text from the images first.

Answer:

Enable prompt shields: Enabled Document attack detection: Enabled

Question: 50

CertyIQ

You are deploying a support agent that enables users to upload photos.

You need to automatically classify uploaded images for harmful content. The solution must block content based on severity levels.

What should you do?

- A. Apply keyword scanning to optical character recognition (OCR) output by using Azure Vision in Foundry Tools.
- B. Enable prompt shields.
- C. Use blocklists.
- D. Implement image moderation.

Answer: D

Explanation:

Technical Justification for Correct Answer: D

Why D (Implement image moderation) is the best option: Implementing image moderation is the most suitable choice for automatically classifying uploaded images for harmful content and blocking based on severity levels. This is because image moderation, specifically through Azure's Computer Vision or Azure Content Moderator services, is designed to analyze visual content. These services can detect and categorize harmful content (e.g., adult, racy, or violent images) with a level of granularity that allows for severity-based blocking. This approach directly addresses the requirement for visual content analysis.

Why other options are less suitable:

A. Apply keyword scanning to optical character recognition (OCR) output by using Azure Vision in Foundry Tools:

Limitation: This method only analyzes text extracted from images via OCR, not the visual content itself. It cannot detect harmful non-textual content (e.g., violence, nudity).

Inappropriateness for Requirement: Fails to directly assess image content for harmful visuals.

B. Enable prompt shields:

Irrelevance: Prompt shields are more relevant to controlling or guiding user input (e.g., in chatbots) rather than analyzing uploaded media for harmful content.

Functional Misalignment: Does not provide a mechanism for image content analysis.

C. Use blocklists:

Limitation: Blocklists are effective for known harmful sources (URLs, IPs) but do not analyze the content of uploaded images in real-time.

Inadequacy for Dynamic Content: Unable to classify new, unseen harmful image content based on its visual characteristics.

References:

[Azure Content Moderator Documentation](#) - For details on moderating images for harmful content.

[Azure Computer Vision Documentation](#) - Although primarily for image analysis, can be used in conjunction with custom logic for moderation purposes.

Question: 51

CertyIQ

You have an app named App1 that uses a Microsoft Foundry multimodal model deployment. App1 runs optical character recognition (OCR) on uploaded images and appends the OCR output to the prompt as additional context.

Some uploaded images contain embedded text.

You need to prevent potentially malicious instructions from being processed by the model.

What should you use?

- A. image moderation
- B. prompt shields for documents
- C. protected material text
- D. prompt shields for user prompts

Answer: B

Explanation:

Technical Justification for Correct Answer: B - Prompt Shields for Documents

Why B is the best option: Prompt Shields for Documents is specifically designed to sanitize and control the input context (in this case, the OCR output appended to the prompt) before it's processed by the model. This feature is particularly relevant when dealing with untrusted or potentially malicious content extracted from documents (like embedded text in images). By using Prompt Shields for Documents, App1 can effectively filter out or redact potentially malicious instructions embedded within the OCR output, preventing them from influencing the model's response.

Why other options are less suitable:

A. Image Moderation: While Image Moderation is crucial for screening out inappropriate or harmful visual content, it does not address the issue of malicious **text** embedded within images. Image Moderation would not process or sanitize the OCR-extracted text, leaving the model vulnerable to malicious instructions in the text output.

C. Protected Material Text: This option seems to imply a feature for handling sensitive or protected text. However, the primary concern here is not the protection of sensitive text but the prevention of malicious text from being processed. Protected Material Text does not inherently imply a mechanism for filtering out malicious instructions, making it less suited for this specific requirement.

D. Prompt Shields for User Prompts: Although similar in name and concept to the correct answer, Prompt Shields for User Prompts is designed to sanitize and control **direct user inputs** (e.g., text typed by users). While it shares the goal of preventing malicious input, it's tailored for user-generated text prompts rather than automatically generated context from documents (like OCR output from images).

Conclusion: Given the specific scenario of preventing potentially malicious **embedded text in images** (after OCR processing) from being processed by the model, **B. Prompt Shields for Documents** is the most appropriate and technically justified choice. It directly addresses the need to sanitize document-extracted text context before model processing.

References

1. **Microsoft Azure - Protecting Models from Malicious Input:** <https://learn.microsoft.com/en-us/azure/cognitive-services/form-recognizer/concept-model-protection> (Though the link focuses on Form Recognizer, the concept of protecting models from malicious input is broadly applicable and relevant to understanding the importance of sanitizing inputs.)
2. **Microsoft Documentation - Azure Cognitive Services Security and Compliance:** <https://learn.microsoft.com/en-us/azure/cognitive-services/security-compliance> (This link provides a broad overview of security measures in Azure Cognitive Services, including input validation and sanitization principles that underpin the rationale for using Prompt Shields.)

Question: 52

CertyIQ

You have a Microsoft Foundry project that contains an agent.

You need to process mixed-format documents that contain scanned text, tables, and multicolumn layouts. The extracted content must preserve the document structure and be converted into the Markdown format for downstream reasoning.

What should you configure first?

- A. an Azure Language in Foundry Tools text analysis model deployment
- B. a generative chat completion request
- C. an Azure OpenAI Responses API call that uses a multimodal model
- D. an Azure Content Understanding in Foundry Tools analyzer

Answer: D

Explanation:

Technical Justification for Correct Answer: D

To address the requirement of processing mixed-format documents (containing scanned text, tables, and multicolumn layouts) while preserving the document structure and converting the extracted content into Markdown format for downstream reasoning, the most appropriate initial configuration is **D. an Azure Content Understanding in Foundry Tools analyzer**. Here's why:

Why D is the Correct Choice:

Document Structure Preservation: Azure Content Understanding is designed to comprehend and retain the structural integrity of documents, including tables and multicolumn layouts, which is crucial for this task.

Support for Mixed-Format Documents: It can handle scanned documents (via integration with OCR capabilities implicitly or explicitly through associated services) alongside other formats, making it suitable for mixed-format input.

Output Flexibility: The extracted, structured content can be more easily converted into Markdown format due to its understanding and output of document components (e.g., headings, paragraphs, tables).

Why Other Options are Less Suitable:

A. Azure Language in Foundry Tools text analysis model deployment:

Limitation: Primarily focused on text analysis (sentiment, entity recognition, etc.) rather than preserving document structure or handling scanned/multicolumn content effectively.

Conversion to Markdown: Would require additional, separate processing to structure the output into Markdown, increasing the complexity of the pipeline.

B. a generative chat completion request:

Purpose Misalignment: Designed for generating human-like text based on prompts, not for document structure analysis or preservation.

Format Conversion: Not suited for accurately converting structured document content into Markdown.

C. an Azure OpenAI Responses API call that uses a multimodal model:

Multimodal Capability: While powerful, the primary challenge here isn't necessarily the type of input (though multimodal helps), but accurately extracting and preserving document structure.

Structure Preservation and Markdown Conversion: Less directly suited for the precise structural analysis and conversion requirements compared to a content understanding tool.

Cost and Complexity: Might introduce unnecessary complexity and cost for the specific task at hand, given the more targeted capability of Content Understanding.

Conclusion:

Given the specific requirements of preserving document structure, handling mixed formats (including scanned documents), and converting to Markdown, **configuring an Azure Content Understanding in Foundry Tools analyzer first** is the most efficient and technically appropriate approach.

References:

Question: 53

You have an application that processes scanned PDF invoices. The invoices have varied layouts and include multipage tables.

You have a pipeline that uses optical character recognition (OCR) and extracts totals and invoice numbers. The results are often incorrect because the document structure is ignored.

You need to implement a solution that provides OCR, layout analysis, and template-generalizing field extraction. The solution must NOT require training a custom model. The solution must minimize administrative effort. What should you include in the solution?

- A. Azure Language in Foundry Tools
- B. Azure Content Understanding in Foundry Tools
- C. an Azure Machine Learning model

Answer: B

Explanation:

Justification for Correct Answer: B - Azure Content Understanding in Foundry Tools

The correct solution for the given scenario is **B. Azure Content Understanding in Foundry Tools**. Here's a detailed breakdown of why this is the most suitable option and the shortcomings of the other choices:

Why B (Azure Content Understanding in Foundry Tools) is Correct:

OCR, Layout Analysis, and Field Extraction: Azure Content Understanding is designed to handle complex document structures, performing advanced OCR, layout analysis, and field extraction. It can adapt to varied layouts and multipage tables without requiring custom model training.

No Custom Model Training Required: This solution aligns with the requirement of not needing to train a custom model, as it leverages pre-trained models optimized for document understanding.

Minimizes Administrative Effort: By utilizing a managed service, the administrative burden is significantly reduced compared to building and maintaining a custom solution or even managing an Azure Machine Learning model.

Why Other Options are Less Suitable:

A. Azure Language in Foundry Tools:

Primary Focus: More tailored towards natural language processing (NLP) tasks such as text analysis, sentiment analysis, etc., rather than the structured document analysis required for invoices.

Inadequate for Layout Analysis and Template-Generalizing Extraction: Less capable in handling the complexities of varying document layouts and the specific need for accurate field extraction from tables and forms.

C. An Azure Machine Learning Model:

Requires Custom Model Training: Directly contradicts the requirement of not needing to train a custom model, which would also increase development time and effort.

Higher Administrative Effort: Managing, updating, and ensuring the model's accuracy over time would demand more administrative resources compared to a managed service like Azure Content Understanding.

References

For further reading and to delve deeper into the capabilities of the recommended solution:

1. **Azure Content Understanding Documentation** - <https://learn.microsoft.com/en-us/azure/foundry-tools/content-understanding/>
2. **Comparison of Azure Services for Document Processing** - <https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/document-processing-services> (Utilize this to compare the functionalities of different Azure services for document processing tasks.)

Question: 54

CertyIQ

You have a Microsoft Foundry project that contains an agent.

The agent uses a knowledge source built from documents stored in Azure Blob Storage. The documents include digitally scanned PDFs that contain multipage tables.

You have an ingestion job that extracts only plain text, causing loss of table structure, headings, and page-number metadata.

Users frequently ask questions that require the retrieval of specific table rows across the pages.

You need to configure an ingestion job for a Retrieval Augmented Generation (RAG) pipeline that performs optical character recognition (OCR) on scanned PDFs, preserves tables and headings as structure-aware chunks, and stores page-number metadata with each chunk.

How should you configure the ingestion job?

- A. Use advanced data parsing to reingest the documents.
- B. Use OCR and page-level chunking.
- C. Use page-level OCR extraction and store each page as a single chunk.
- D. Use basic parsing and fixed-size chunking.

Answer: A

Explanation:

Technical Justification for Correct Option (A)

The correct configuration for the ingestion job, given the requirements, is **A. Use advanced data parsing to reingest the documents**. Here's why this option stands out and why others are less suitable:

Why A is the Best Choice:

Advanced Data Parsing Capability: Advanced data parsing is specifically designed to handle complex document structures, including the preservation of tables, headings, and extraction of specific metadata like page numbers. This aligns perfectly with the need to retain the structure of multipage tables and associate page-number metadata with each chunk.

Retrieval Augmented Generation (RAG) Pipeline Compatibility: For a RAG pipeline, the ingestion job needs to provide structured, searchable content. Advanced data parsing ensures that the output is optimized for such pipelines, facilitating efficient retrieval of specific table rows across pages.

OCR Implicit in Requirement Understanding: While not explicitly mentioned in option A, the context implies that OCR is necessary for scanned PDFs. Advanced data parsing tools often integrate or can be configured with OCR capabilities to process scanned documents, making this a holistic choice for the given scenario.

Why Other Options are Less Suitable:

B. Use OCR and Page-Level Chunking:

Lack of Explicit Structure Preservation: There's no clear indication that this option preserves table structures and headings as "structure-aware chunks" beyond basic page-level processing.

Insufficient for RAG's Structured Needs: Might not fully support the nuanced search and retrieval requirements of a RAG pipeline without advanced parsing.

C. Use Page-Level OCR Extraction and Store Each Page as a Single Chunk:

Overly Broad Chunks: Storing each page as a single chunk does not facilitate the retrieval of specific table rows efficiently, as it doesn't preserve the granular structure of the content within pages.

Metadata Limitation: Implicitly suggests a lack of detailed metadata association (like specific table row locations) with each chunk.

D. Use Basic Parsing and Fixed-Size Chunking:

Incompatible with Complex Structures: Basic parsing cannot preserve the complex table structures and headings effectively.

Fixed-Size Chunking Inefficiency: Irrespective of content structure, fixed-size chunking is inefficient for retrieving specific, variable-sized content elements (like table rows) and does not account for page-number metadata.

References

[Microsoft Azure - Cognitive Search: Document AI](#)

[Azure Cognitive Services - Form Recognizer for Structured Data Extraction](#)

Question: 55

CertyIQ

You have a Microsoft Foundry project that contains an agent.

The agent uses Azure AI Search as the retriever.

You plan to ingest PDF into an Azure AI Search index to ensure that the agent can ground responses in texts in both documents and embedded images.

Users require citations that link to the source files.

You need to ensure that during indexing, the images are extracted into a structure that can be used as input for the built-in optical character recognition (OCR) skill.

Which indexing approach should you use?

- A. an indexer to extract image data into a normalized_images collection
- B. a Shaper skill to restructure the OCR input
- C. a skillset to run the OCR skill directly against the content field of the index
- D. the outputFieldMappings parameter to write image data to a searchable field

Answer: A

Explanation:

Technical Justification for Correct Answer: A

The correct indexing approach to ensure images are extracted into a structure suitable for the built-in Optical Character Recognition (OCR) skill, while also supporting user requirements for citations linking to source files, is:

A. an indexer to extract image data into a normalized_images collection

Why A is the Best Choice:

Direct Extraction and Structuring: Using an indexer to extract image data into a normalized_images collection directly structures the image data in a way that is readily consumable by the OCR skill, streamlining the

process.

Support for Citations: By keeping the image data linked to its source (through the structured collection), it facilitates the creation of citations that can point back to the original PDF files, meeting the user's requirement.

Efficiency and Scalability: Indexers are designed for such data ingestion and transformation tasks, making this approach efficient and scalable for handling multiple PDFs and their embedded images.

Why Other Options are Less Suitable:

B. a Shaper skill to restructure the OCR input:

Inappropriate Timing: Shaper skills are used for reshaping data after processing (e.g., post-OCR), not for pre-processing image extraction.

Inefficient for Image Extraction: Not designed for the initial extraction of images from documents.

C. a skillset to run the OCR skill directly against the content field of the index:

Lack of Image Extraction: Running OCR directly against the content field does not address the extraction of images from PDFs into a usable structure.

Missed Structuring Opportunity: Fails to preprocess images into an optimal structure for OCR, potentially leading to less accurate results.

D. the outputFieldMappings parameter to write image data to a searchable field:

Insufficient Structuring for OCR: While useful for making data searchable, it does not ensure the image data is structured appropriately for effective OCR processing.

Citations and Source Linking: Less directly supports the maintenance of clear links to source files for citations compared to a dedicated collection approach.

References

[Microsoft Azure Documentation: Azure Cognitive Search Indexers](#)

[Microsoft Azure Documentation: Skillsets in Azure Cognitive Search](#)

Question: 56

CertyIQ

Case Study -

This is a case study. Case studies are not timed separately from other exam sections. You can use as much exam time as you would like to complete each case study. However, there might be additional case studies or other exam sections. Manage your time to ensure that you can complete all the exam sections in the time provided. Pay attention to the Exam Progress at the top of the screen so you have sufficient time to complete any exam sections that follow this case study.

To answer the case study questions, you will need to reference information that is provided in the case. Case studies and associated questions might contain exhibits or other resources that provide more information about the scenario described in the case. Information provided in an individual question does not apply to the other questions in the case study.

A Review Screen will appear at the end of this case study. From the Review Screen, you can review and change your answers before you move to the next exam section. After you leave this case study, you will NOT be able to return to it.

To start the case study -

To display the first question in this case study, select the "Next" button. To the left of the question, a menu provides links to information such as business requirements, the existing environment, and problem statements. Please read through all this information before answering any questions. When you are ready to answer a question, select the "Question" button to return to the question.

Overview -

Company Information -

Contoso, Ltd is a multinational retail company that builds, deploys, and manages generative AI and agent-based solutions by using Microsoft Foundry.

Existing Environment -

Identity Environment -

Contoso uses Microsoft Entra ID for identity management, authentication, and authorization capabilities that enable agents to access organizational resources and services.

Contoso recently formed a new AI engineering team named Agent1Dev Team to optimize and maintain existing AI solutions.

The team collaborates with solution architects, DevOps engineers, and security engineers to design, implement, monitor, and secure AI applications.

Contoso also has a team named Agent1Test Team that is responsible for validating AI solutions before the solution deployments.

Generative Environment -

Contoso has a Microsoft Foundry deployment that contains two projects named Project1 and Project2.

Project1 -

Project1 contains a customer support agent named Agent1 that assists customers with product inquiries and troubleshooting requests.

Agent1 has the following configurations:

Agent1 uses a base model deployment.

A safety evaluation pipeline is NOT enabled.

Tool invocation approval workflows are NOT enabled.

Conversation memory constraints are NOT configured.

Agent1 interacts with customers by using digital support channels and answers general questions about Contoso products.

Project1 is deployed to an Azure region located in the European Union (EU).

Agent1Dev Team will use Project1 to optimize and maintain Agent1.

Project2 -

Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.

Development of the solution is incomplete.

Data Environment -

Contoso stores product-related information in Azure resources that support AI applications.

The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.

The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.

Problem Statements -

Contoso identifies the following issues:

Agent1 has only general knowledge of the Contoso products.

A recent chat interaction with Agent1 was analyzed for sentiment. The results of the analysis have NOT been processed yet.

Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.

The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.

Requirements -

Planned Changes -

Contoso plans to implement the following changes:

Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.

Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.

Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.

Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.

Complete the development of the video creation solution.

Technical Requirements -

Contoso identifies the following technical requirements:

The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.

The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.

Responses generated by using the product sheet information must be relevant, complete, and accurate.

Agent1 must be able to use the product sheets to answer natural language questions about product details.

The model version used by Agent1 must remain consistent to ensure stable responses.

The data processed by the model must remain within the EU.

Security and Compliance Requirements

Contoso identifies the following security and compliance requirements:

API keys must NOT be used to access Foundry-deployed models.

Access to the Azure resources must follow the principle of least privilege.

The developers at Contoso must authenticate to Microsoft Foundry resources by using Microsoft Entra authentication.

Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.

Access to Project1 must be assigned to the members of Agent1Test Team by using a security group named SC_Agent1_Test.

Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.

The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.

Business Requirements -

Contoso identifies the following business requirements:

Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.

Agent1 must answer questions only about the products sold by Contoso.

You need to recommend an invoice review solution that resolves the issue reported by the finance department.

What should you include in the recommendation?

- A. chat completions
- B. Azure Document Intelligence in Foundry Tools
- C. Azure Content Understanding in Foundry Tools
- D. Image Analysis

Answer: C

Explanation:

Technical Justification for Invoice Review Solution Recommendation

Correct Option: C. Azure Content Understanding in Foundry Tools

Azure Content Understanding in Foundry Tools is the most suitable choice for the invoice review solution due to the following technical reasons:

Visual and Textual Content Evaluation: The finance department's issue involves verifying invoice details against vendor contract terms, which includes analyzing both the visual layout (e.g., tables, logos, varied layouts) and the textual content of the invoices. Azure Content Understanding is adept at handling both aspects, providing a comprehensive analysis.

Integration with Foundry: Since Contoso already utilizes Microsoft Foundry, integrating Azure Content Understanding ensures seamless interaction with existing infrastructure, streamlining the development and deployment process.

Scalability and Throughput: While not explicitly mentioned for this specific task, the overall technical requirements for Contoso's AI solutions emphasize scalability and high-throughput capabilities, which Azure Content Understanding can support, especially when combined with other Azure services optimized for such workloads.

Why Other Options Are Less Suitable:

A. Chat Completions:

Inappropriate Use Case: Chat completions are more suited for generating text based on partial inputs, not for analyzing and verifying document content against external terms.

Lack of Visual Analysis Capability: This option does not address the visual layout analysis requirement.

B. Azure Document Intelligence in Foundry Tools:

More General-Purpose: While capable, Azure Document Intelligence might not offer the depth of content

understanding required for nuanced invoice verification, especially if the focus is more on the intelligence extracted from the content rather than just its structure.

Assumes Structured Data Extraction Needs: The primary issue isn't just extracting structured data but understanding and verifying it against contract terms, which might require the deeper content analysis capabilities of Azure Content Understanding.

D. Image Analysis:

Too Narrow in Scope: Image Analysis would handle the visual aspects and potentially images with embedded text but would not comprehensively address the textual content analysis and verification against contract terms without additional, unnecessary integration complexity.

References:

For further details on the recommended solution and its capabilities:

1. **Azure Content Understanding Documentation:** <https://learn.microsoft.com/en-us/azure/cognitive-services/content-understanding/overview>
2. **Microsoft Foundry Integration with Azure Services:** <https://learn.microsoft.com/en-us/azure/foundry/how-to-integrate-azure-services>

Question: 57

CertyIQ

You have a Microsoft Foundry project that contains an agent.

The knowledge source for the agent is a set of scanned PDF troubleshooting guides stored in Azure Blob Storage. The guide pages contain two-column layouts and tables.

You use Azure Content Understanding in Foundry Tools to process the PDFs.

You plan to ingest the processed content into an index for Retrieval Augmented Generation (RAG) and store extracted fields for downstream automation.

Stakeholders must be able to verify where each extracted field value came from in the original PDF and route low-reliability extractions for manual review.

You need to ensure that the Content Understanding document analyzer output includes a per-field confidence score and source grounding to locations within the source document.

What should you do?

- A. Set enableSegment to true.
- B. Provide labeled samples.
- C. Enable estimateFieldSourceAndConfidence.
- D. Configure the analyzer to use generative extraction for all fields.

Answer: C

Explanation:

Technical Justification for Correct Answer: C

To address the requirements of including per-field confidence scores and source grounding to locations within the source document for the Azure Content Understanding document analyzer output in the context of a Microsoft Foundry project, **enabling estimateFieldSourceAndConfidence (Option C)** is the most appropriate action. Here's why:

Per-field Confidence Score Requirement: EstimateFieldSourceAndConfidence is designed to not only estimate the confidence in each extracted field value but also to identify the source location within the document where the field value was extracted from. This directly fulfills the need for per-field confidence scores, ensuring stakeholders can assess the reliability of each extraction.

Source Grounding Requirement: By enabling this feature, the output will include references to the exact locations in the original PDF where each field's value was sourced. This feature is crucial for traceability, especially in documents with complex layouts like two-column setups and tables, as it facilitates verification of the extraction's origin.

Why Other Options are Less Suitable:

A. Set enableSegment to true: While enabling segmentation can improve the processing of structured content by breaking down documents into manageable parts, it does not directly address the need for confidence scores or source grounding for extracted fields. Segmentation is more about the preprocessing of the document rather than the extraction metadata.

B. Provide labeled samples: Providing labeled samples is essential for training and improving the accuracy of the model over time. However, this action does not directly influence the inclusion of confidence scores or source information in the analyzer's output for immediate processing needs. It's a long-term solution for model enhancement rather than a direct fix for the specified requirements.

D. Configure the analyzer to use generative extraction for all fields: Generative extraction can be powerful for certain types of data extraction tasks, especially where the model needs to generate text based on the document content. However, this approach may not always provide the precise source grounding or confidence scores per field as directly and reliably as enabling EstimateFieldSourceAndConfidence, especially in the context of the described requirements and the structured nature of the data (tables, two-column layouts).

Correct Action: C. Enable estimateFieldSourceAndConfidence

Enabling this feature ensures that the Content Understanding document analyzer output meets both critical requirements (per-field confidence scores and source grounding) efficiently and directly, making it the best choice for the given scenario.

References

1. **Azure Cognitive Search Documentation - Content Understanding:** <https://docs.microsoft.com/en-us/azure/search/cognitive-search-concept-intro#content-understanding>
2. **Customizing Document Analyzer for Content Understanding:** <https://docs.microsoft.com/en-us/azure/search/cognitive-search-defining-skillset#document-analyzer-skills> (Refer to sections related to field extraction and confidence scores for more detailed insights)

Question: 58

CertyIQ

You are building a speech processing solution in Microsoft Foundry for a customer support platform. The platform will transcribe live phone calls, so that supervisors at your company can view call transcripts and detect issues while the calls are in progress. The call audio will arrive as a continuous stream from the telephony system.

You need to ensure that the call transcripts appear within only a few seconds of the audio stream. What should you do?

- A. Use text to speech by using a custom neural voice.
- B. Use speech translation to generate the transcripts into multiple languages.
- C. Run a batch transcription job on recorded audio files.
- D. Use real-time speech to text to process streaming audio input.

Answer: D

Explanation:

Technical Justification for the Correct Option (D)

To meet the requirement of transcribing live phone calls with near-real-time transcription (within a few seconds of the audio stream), the solution must process the audio input as a continuous stream rather than in batches or after the call is completed. Here's why option D is the best choice and why the others are less suitable:

Correct Option: D - Use real-time speech to text to process streaming audio input

Real-time Capability: Real-time speech to text is designed to handle streaming audio inputs, making it perfectly suited for transcribing live phone calls.

Low Latency: This approach ensures that transcripts are generated with minimal delay, fulfilling the requirement of transcripts appearing within a few seconds of the audio stream.

Continuous Streaming Support: It is specifically built to process continuous audio streams, such as those from ongoing telephony systems, without requiring the audio to be pre-recorded or buffered for an excessive amount of time.

Why Other Options Are Less Suitable

A - Use text to speech by using a custom neural voice

Reverse Requirement: Text to speech converts text into speech, which is the opposite of what is needed (converting speech to text).

No Benefit for Transcription: Custom neural voices do not aid in the transcription process but rather in generating synthetic speech from text.

B - Use speech translation to generate the transcripts into multiple languages

Additional, Unrequested Functionality: While useful for multi-language support, the primary requirement is for real-time transcription in one language (implied, as the question does not mention translation needs).

Potential Increase in Latency: Adding a translation step could introduce additional delay, moving further from the near-real-time requirement.

C - Run a batch transcription job on recorded audio files

Batch Processing Incompatibility: Batch jobs are designed for processing pre-recorded audio files in bulk, not for real-time or near-real-time transcription of live streams.

High Latency: Transcripts would only be available after the batch job completes, which would be long after the call has ended, violating the low latency requirement.

Question: 59

CertyIQ

You are creating an agent workflow in a Microsoft Foundry project to support natural voice interactions. The agent must receive continuous audio input, convert the input into text for reasoning, and then return spoken responses to a user. The workflow must meet the following requirements:

Support turn-taking dynamics, where the agent begins to generate the speech output before the user finishes speaking.

Operate with low latency to maintain conversational experience.

You need to enable both speech to text and text to speech in a real-time agent interaction.

What should you do?

- A. Use batch transcription to convert the audio input and return text responses from the agent.
- B. Use real-time speech to text for incoming audio and text to speech for agent responses.
- C. Use an embeddings model to encode the audio, and then decode the audio into text and speech.

D. Use speech translation to convert the audio into another language and return the translated text.

Answer: B

Explanation:

Technical Justification for Correct Answer: B

Why B is the Best Option:

Real-time Processing: Option B utilizes real-time speech to text, which is essential for meeting the low latency requirement, ensuring a seamless conversational experience. This capability processes audio as it is being received, minimizing delays.

Support for Turn-Taking Dynamics: By converting audio to text in real-time, the agent can begin generating speech output even before the user finishes speaking, directly supporting the turn-taking dynamics requirement.

Direct Alignment with Requirements:

Speech to Text: Directly converts incoming audio into text for the agent's reasoning, fulfilling the first technical requirement.

Text to Speech: Generates spoken responses from the text output, fulfilling the second technical requirement for real-time agent interaction.

Why Other Options are Less Suitable:

A. Batch Transcription:

High Latency: Processes audio in batches, leading to significant delays, which contradicts the low latency requirement.

Incompatible with Turn-Taking: Cannot start processing until the user finishes speaking, making turn-taking dynamics impossible.

C. Embeddings Model for Encoding/Decoding:

Indirect Approach: While embeddings can be used in NLP tasks, this option describes an indirect and potentially less efficient method for the specific tasks of real-time speech to text and text to speech conversion.

Lack of Clear Latency Benefit: Does not inherently guarantee low latency or support for turn-taking as effectively as dedicated real-time speech services.

D. Speech Translation:

Misaligned Capability: Focuses on translating audio into another language, which is not the primary requirement (conversational interaction in the same language).

Added Latency: Translation adds an extra step, potentially increasing latency beyond what's acceptable for real-time interaction.

Conclusion: Given the requirements for real-time processing, low latency, support for turn-taking dynamics, and the direct need for speech to text and text to speech conversion, **Option B** is the most technically suitable choice for enabling both functionalities in a real-time agent interaction.

References

1. **Microsoft Azure Speech Services - Real-time Speech Recognition**<https://docs.microsoft.com/en-us/azure/cognitive-services/speech-service/speech-to-text>
2. **Microsoft Azure Text-to-Speech (TTS) for Real-time Responses**<https://docs.microsoft.com/en-us/azure/cognitive-services/speech-service/text-to-speech>

Question: 60

You have an application named App1 that uses Azure Speech in Foundry Tools to transcribe live calls. Transcript segments often contain both English and Spanish. App1 sends each segment to Azure Translator in Foundry Tools to translate to another language.

Sometimes, mixed-language segments result in incomplete or incorrect translations.

You need to reduce translation errors. The solution must ensure that the entire transcript is translated successfully.

What should you do before sending the segments to Translator?

- A. Use document translation to translate the entire transcript as a single document.
- B. Split the mixed-language segments into single-language segments and translate each segment separately.
- C. Enable automatic language detection for the translation request.
- D. Specify English as the source language in the translation request for all the segments.

Answer: B

Explanation:

Technical Justification for Correct Answer: B

To address the issue of incomplete or incorrect translations for mixed-language segments in App1, the most effective solution is to **split the mixed-language segments into single-language segments and translate each segment separately (Option B)**. Here's why this approach is best and why other options are less suitable:

Why B is the Correct Approach:

Accuracy Enhancement: By isolating language segments, the translation model can focus on one language at a time, significantly improving accuracy.

Direct Handling of Mixed-Language Content: This method proactively deals with the root cause (mixed languages in a segment) by separating them, ensuring each translation request is linguistically homogeneous.

Compatibility with Azure Translator: Azure Translator can efficiently handle multiple, single-language translation requests, making this approach technically feasible and scalable.

Why Other Options are Less Suitable:

A. Use Document Translation for the Entire Transcript:

Limitation: While document translation can handle larger texts, it may not effectively mitigate mixed-language issues within a single document, as the engine might struggle with interwoven languages.

Potential for Errors: Similar or worse error rates as current setup if language detection within the document feature is not robust enough for intertwined languages.

C. Enable Automatic Language Detection for the Translation Request:

Reliability Concerns: Automatic detection may not always accurately identify and separate mixed languages within a single segment, potentially leading to similar translation errors.

Segmentation Challenge: Detection doesn't solve the segmentation issue; it only identifies languages, not necessarily how to process them correctly in mixed segments.

D. Specify English as the Source Language for All Segments:

Inapplicability: Ignores the presence of Spanish (and potentially other languages), guaranteeing translation

failures for non-English parts.

Error Rate Increase: Likely to increase errors for segments containing languages other than English.

Conclusion: Splitting mixed-language segments into single-language parts before translation (Option B) is the most technically sound approach to reduce translation errors, given its direct addressing of the problem's root cause and compatibility with the capabilities of Azure Translator in Foundry Tools.

References

1. **Azure Translator Documentation - Best Practices for Translation Requests:**
<https://docs.microsoft.com/en-us/azure/cognitive-services/translator/best-practices>
2. **Azure Speech to Translator Integration Guidance:** <https://docs.microsoft.com/en-us/azure/cognitive-services/speech-service/howto-translate-speech?tabs=dotnet> (Note: While this link focuses on speech-to-text-to-speech translation, the principles of handling multi-language inputs can be inferred for text translation scenarios.)

Question: 61

CertyIQ

Case Study -

This is a case study. Case studies are not timed separately from other exam sections. You can use as much exam time as you would like to complete each case study. However, there might be additional case studies or other exam sections. Manage your time to ensure that you can complete all the exam sections in the time provided. Pay attention to the Exam Progress at the top of the screen so you have sufficient time to complete any exam sections that follow this case study.

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A Review Screen will appear at the end of this case study. From the Review Screen, you can review and change your answers before you move to the next exam section. After you leave this case study, you will NOT be able to return to it.

To start the case study -

To display the first question in this case study, select the "Next" button. To the left of the question, a menu provides links to information such as business requirements, the existing environment, and problem statements. Please read through all this information before answering any questions. When you are ready to answer a question, select the "Question" button to return to the question.

Overview -

Company Information -

Contoso, Ltd is a multinational retail company that builds, deploys, and manages generative AI and agent-based solutions by using Microsoft Foundry.

Existing Environment -

Identity Environment -

Contoso uses Microsoft Entra ID for identity management, authentication, and authorization capabilities that enable agents to access organizational resources and services.

Contoso recently formed a new AI engineering team named Agent1Dev Team to optimize and maintain existing AI solutions.

The team collaborates with solution architects, DevOps engineers, and security engineers to design, implement, monitor, and secure AI applications.

Contoso also has a team named Agent1Test Team that is responsible for validating AI solutions before the solution deployments.

Generative Environment -

Contoso has a Microsoft Foundry deployment that contains two projects named Project1 and Project2.

Project1 -

Project1 contains a customer support agent named Agent1 that assists customers with product inquiries and troubleshooting requests.

Agent1 has the following configurations:

Agent1 uses a base model deployment.

A safety evaluation pipeline is NOT enabled.

Tool invocation approval workflows are NOT enabled.

Conversation memory constraints are NOT configured.

Agent1 interacts with customers by using digital support channels and answers general questions about Contoso

products.

Project1 is deployed to an Azure region located in the European Union (EU).

Agent1Dev Team will use Project1 to optimize and maintain Agent1.

Project2 -

Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.

Development of the solution is incomplete.

Data Environment -

Contoso stores product-related information in Azure resources that support AI applications.

The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.

The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.

Problem Statements -

Contoso identifies the following issues:

Agent1 has only general knowledge of the Contoso products.

A recent chat interaction with Agent1 was analyzed for sentiment. The results of the analysis have NOT been processed yet.

Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.

The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.

Requirements -

Planned Changes -

Contoso plans to implement the following changes:

Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.

Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.

Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.

Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.

Complete the development of the video creation solution.

Technical Requirements -

Contoso identifies the following technical requirements:

The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.

The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.

Responses generated by using the product sheet information must be relevant, complete, and accurate.

Agent1 must be able to use the product sheets to answer natural language questions about product details.

The model version used by Agent1 must remain consistent to ensure stable responses.

The data processed by the model must remain within the EU.

Security and Compliance Requirements

Contoso identifies the following security and compliance requirements:

API keys must NOT be used to access Foundry-deployed models.

Access to the Azure resources must follow the principle of least privilege.

The developers at Contoso must authenticate to Microsoft Foundry resources by using Microsoft Entra authentication.

Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.

Access to Project1 must be assigned to the members of Agent1Test Team by using a security group named SC_Agent1_Test.

Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.

The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.

Business Requirements -

Contoso identifies the following business requirements:

Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.

Agent1 must answer questions only about the products sold by Contoso.

You need to configure an indexing pipeline for Agent1 to retrieve the relevant product information in storage1. The solution must meet the technical requirement.

Which two built-in skills should you use? Each correct answer presents part of the solution.

NOTE: Each correct selection is worth one point.

- A. Azure OpenAI Embedding
- B. Entity Recognition
- C. Text Split
- D. Merge
- E. Language Detection
- F. key phrase extraction

Answer: AC

Explanation:

Technical Justification for Correct Answer: AC (Azure OpenAI Embedding & Text Split)

Contoso requires an indexing pipeline for Agent1 to retrieve relevant product information from PDF product sheets in storage1, meeting the technical requirement of enabling semantic and vector search for accurate, complete, and relevant responses to natural language questions.

Why Azure OpenAI Embedding (A) is Correct:

Semantic and Vector Search Capability: Azure OpenAI Embedding is designed to generate dense vector representations of text, enabling efficient semantic search. This aligns perfectly with Contoso's requirement for an indexing pipeline that facilitates semantic and vector search, allowing Agent1 to retrieve relevant product information based on the semantic meaning of the queries.

Integration with AI Workloads: Given Contoso's use of Microsoft Foundry for generative AI solutions, Azure OpenAI Embedding integrates seamlessly, supporting the scalable, high-throughput requirements for Agent1's interactions.

Handling Natural Language Queries: By generating embeddings, this skill enhances Agent1's ability to understand and respond accurately to natural language questions about product details, ensuring relevance and completeness.

Why Text Split (C) is Correct:

Processing Large Documents: Product sheets are in PDF format, which can be lengthy. Text Split is essential for breaking down these large documents into manageable, indexing-friendly segments, ensuring that the indexing pipeline can efficiently process each part.

Optimizing Search Efficiency: By splitting text, the search index can be more granular, leading to faster and more accurate retrieval of specific product information in response to customer queries.

Preprocessing for Embedding: Text Split prepares the text for more effective embedding generation by Azure OpenAI Embedding, as smaller, contextually relevant chunks of text can produce more accurate vector representations.

Why Other Options are Less Suitable:

B. Entity Recognition: While useful for identifying specific entities (e.g., product names), it does not enable the broad semantic and vector search required for retrieving relevant product information based on natural language queries.

D. Merge: This skill is counterproductive for the goal of creating a granular, searchable index, as it combines texts, potentially reducing search efficiency.

E. Language Detection: Not relevant to the task of indexing product information for semantic/vector search, as the language of the product sheets is presumably known and consistent.

F. Key Phrase Extraction: Similar to Entity Recognition, it's useful but doesn't meet the requirement for a searchable index that supports semantic and vector searches for accurate responses to varied natural

language questions.

References

1. **Azure OpenAI Embedding Documentation:** <https://learn.microsoft.com/en-us/azure/cognitive-services/OpenAI/how-to/embeddings>
2. **Azure Cognitive Search (for context on Text Split and semantic search capabilities):** <https://learn.microsoft.com/en-us/azure/search/search-what-is-azure-search>

Question: 62

CertyIQ

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Project1 is deployed to an Azure region located in the European Union (EU).

Agent1Dev Team will use Project1 to optimize and maintain Agent1.

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Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.

Development of the solution is incomplete.

Data Environment -

Contoso stores product-related information in Azure resources that support AI applications. The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.

The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.

Problem Statements -

Contoso identifies the following issues:

Agent1 has only general knowledge of the Contoso products.

A recent chat interaction with Agent1 was analyzed for sentiment. The results of the analysis have NOT been processed yet.

Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.

The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.

Requirements -

Planned Changes -

Contoso plans to implement the following changes:

Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.

Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.

Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.

Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.

Complete the development of the video creation solution.

Technical Requirements -

Contoso identifies the following technical requirements:

The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.

The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.

Responses generated by using the product sheet information must be relevant, complete, and accurate.

Agent1 must be able to use the product sheets to answer natural language questions about product details.

The model version used by Agent1 must remain consistent to ensure stable responses.

The data processed by the model must remain within the EU.

Security and Compliance Requirements

Contoso identifies the following security and compliance requirements:

API keys must NOT be used to access Foundry-deployed models.

Access to the Azure resources must follow the principle of least privilege.

The developers at Contoso must authenticate to Microsoft Foundry resources by using Microsoft Entra authentication.

Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.

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Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.

The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.

Business Requirements -

Contoso identifies the following business requirements:

Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.

Agent1 must answer questions only about the products sold by Contoso.

You need to recommend a solution to support the planned changes and technical requirements for Agent1 to use the product information stored in storage1.

What should you include in the recommendation?

- A. Azure Translator in Foundry Tools
- B. Grounding with Bing Search
- C. Azure AI Search
- D. Azure Document intelligence in Foundry Tools

Answer: C

Explanation:

Technical Justification for Recommending Azure AI Search (Option C)

Why Option C (Azure AI Search) is the Best Choice:

Meets Technical Requirements:

Enables semantic and vector search for indexing product sheets, allowing Agent1 to retrieve relevant information (aligns with the requirement for an indexing pipeline with semantic and vector search capabilities).

Supports scalable, high-throughput workloads suitable for variable customer support traffic without requiring reserved throughput capacity (supports the model deployment requirements for Agent1).

Integration and Compatibility:

Seamless integration with Microsoft Foundry, ensuring consistency with Contoso's existing environment. Facilitates the use of detailed product information from storage1 in responses, enhancing Agent1's capabilities.

Security and Compliance Alignment:

Does not rely on API keys for access, aligning with the ban on API key usage for Foundry-deployed models. Compatible with least privilege access principles and Microsoft Entra authentication for secure access control.

Why Other Options are Less Suitable:

Option A (Azure Translator in Foundry Tools):

Primarily designed for translation tasks, not suited for indexing or retrieving product information based on semantic/vector search requirements.

Does not address the need for Agent1 to understand and respond with detailed product knowledge.

Option B (Grounding with Bing Search):

External search may not guarantee data privacy or compliance with EU data retention requirements (especially since Project1 is deployed in an EU Azure region).

Less integrated with Contoso's existing Microsoft Foundry and Azure environment, potentially introducing complexity.

Option D (Azure Document Intelligence in Foundry Tools):

While useful for extracting information from documents (like invoices), it doesn't fulfill the semantic/vector search indexing requirement for enabling Agent1 to answer product questions effectively.

More focused on document processing rather than search and retrieval for conversational AI use cases.

Conclusion: Azure AI Search is the most comprehensive solution for Contoso's requirements, offering the necessary search capabilities, scalability, security, and integration with their existing Microsoft Foundry environment to enhance Agent1's functionality effectively.

References:

1. **Azure AI Search Documentation** - <https://learn.microsoft.com/en-us/azure/search/search-what-is-azure-search>
2. **Microsoft Foundry Integration with Azure Services** - <https://learn.microsoft.com/en-us/azure/foundry/how-to-integrate-with-azure-services>

Question: 63

You have a Microsoft Foundry project named Project1. Project1 contains an application that processes PDF vendor invoices. You need to configure Azure Document Intelligence in Foundry Tools to generate a Markdown output that preserves the sections and table structure of the PDFs. The solution must minimize development effort. What should you do?

- A. Configure output=figures when you analyze the PDF.
- B. Configure content=markdown when you analyze the document.
- C. Increase the confidence threshold.
- D. Set the output_content_format=ContentFormat.MARKDOWN value.

Answer: D

Explanation:

Technical Justification for Correct Answer (D)

To configure Azure Document Intelligence in Foundry Tools for generating Markdown output that preserves the sections and table structure of PDF vendor invoices with minimal development effort, the most appropriate action is to **Set the output_content_format=ContentFormat.MARKDOWN value**. Here's why:

Why D is the Correct Choice

Direct Specification of Output Format: By setting output_content_format=ContentFormat.MARKDOWN, you directly instruct the Azure Document Intelligence API to produce the output in Markdown format. This ensures that the structure (including sections and tables) is preserved in a format that's easily readable and further processable.

Minimal Development Effort: This approach requires no additional coding to parse or transform the output, as the API handles the conversion to Markdown. This aligns perfectly with the requirement to minimize development effort.

Why Other Options are Less Suitable

A. Configure output=figures:

Inadequate for Structural Preservation: Specifying output=figures would primarily focus on extracting images or figures from the PDF, neglecting the textual content and structural elements (sections, tables) that need preservation.

Does Not Generate Markdown: This option does not generate Markdown output, failing to meet the primary requirement.

B. Configure content=markdown:

Misinterpretation of Parameter: While this option seems relevant due to the mention of "markdown", the content parameter typically refers to what content to extract (e.g., text, tables) rather than the output format. The correct approach for specifying the output format is through output_content_format.

May Not Preserve Structure as Required: Even if content=markdown influenced output format (which it doesn't directly), there's no guarantee it would preserve structures as effectively as specifying the output format explicitly.

C. Increase the confidence threshold:

Irrelevant to Output Format: Adjusting the confidence threshold affects the accuracy and reliability of the

extracted data (making the model more conservative in what it outputs as recognized text or structures) but has no bearing on the output format or structural preservation in Markdown.

Conclusion

Setting `output_content_format=ContentFormat.MARKDOWN` is the most direct, efficient, and effective method to achieve the desired outcome with minimal development effort, making **D** the correct choice.

References

[Azure Document Intelligence API Reference - Output Formats](#)

[Microsoft Azure Documentation - Document AI Output Configuration](#)

Question: 64

CertyIQ

You have a Microsoft Foundry project that ingests scanned PDF invoices stored in Azure Blob Storage. Each invoice contains printed fine items and has a table-based layout.

Extracted results are stored as structured JSON and used as grounding data for an agent in a Retrieval Augmented Generation (RAG) solution.

You need to create a single analyzer that meets the following requirements:

Extracts the invoice number, invoice date, vendor name, and total amount across varying templates

Returns confidence scores so that results with confidence below 0.80 can be routed for supervisor review

What should you use?

- A. a Foundry agent that has groundedness guardrails enabled to extract invoice fields and confidence scores
- B. a custom Azure Content Understanding in Foundry Tools analyzer that defines the required fields as the extracted fields and the returned confidence scores for routing
- C. the Azure Content Understanding in Foundry Tools prebuilt-layout analyzer
- D. the Azure Content Understanding in Foundry Tools prebuilt-documentSearch analyzer and `search.score` from the Azure AI Search results for routing

Answer: B

Explanation:

Technical Justification for Choosing Option B

Why B is the Best Choice: A custom Azure Content Understanding (ACU) in Foundry Tools analyzer, as described in option B, is the most suitable choice for this scenario because it directly addresses all specified requirements with precision. By defining the required fields (invoice number, invoice date, vendor name, and total amount) as extracted fields, it ensures targeted data extraction across varying templates. Moreover, configuring the analyzer to return confidence scores enables the crucial functionality of routing results with scores below 0.80 for supervisor review, aligning perfectly with the project's needs.

Why Other Options are Less Suitable:

A. Foundry Agent with Groundedness Guardrails:

While a Foundry agent can be powerful for tasks involving grounding data, this option does not directly imply the ability to extract specific fields from scanned PDF invoices with variable templates as efficiently as a customized ACU analyzer.

Confidence score extraction for routing is not a standard feature highlighted by groundedness guardrails, making this option less straightforward for the given requirements.

C. Prebuilt-layout Analyzer in ACU:

The prebuilt-layout analyzer might struggle with "varying templates" since its performance can be highly dependent on the consistency of the layout it's trained on.

Custom extraction of specific fields (invoice number, date, etc.) and the return of confidence scores for routing might not be as directly configurable or as accurate as with a custom analyzer.

D. Prebuilt-documentSearch Analyzer with Azure AI Search Score:

This option focuses more on search functionality rather than precise field extraction from documents, making it less ideal for extracting specific invoice details.

Relying on search.score for confidence might not accurately reflect the extraction confidence of specific fields within the document, as it's more related to the relevance of the document in search results.

Correct Choice Justification Summary:Option B is the most technically appropriate because it allows for:

Customization to extract specific fields across varying templates.

Direct configuration to return confidence scores for conditional routing based on a precise threshold (below 0.80).

References:

[Microsoft Azure Documentation - Custom Analyzers in Content Understanding](#)

[Microsoft Learn - Extract insights from documents with Azure Content Understanding](#)

Question: 65

CertyIQ

You have a Microsoft Foundry project that uses Azure AI Search to ground an agent in internal documentation. After a recent content update, users report that the agent's answers have become less accurate. You need to identify whether the retrieved content is negatively influencing the model's generated responses. Which observability signal should you review?

- A. indexer status and failure history
- B. latency breakdown traces
- C. prediction drift metrics
- D. groundedness evaluation metrics

Answer: D

Explanation:

Technical Justification for Correct Answer (D)

To address the issue of decreased accuracy in the agent's responses after a content update in a Microsoft Foundry project utilizing Azure AI Search for grounding, it's crucial to focus on observability signals that directly relate to the integration of retrieved content with the model's response generation. Here's why **D. Groundedness Evaluation Metrics** is the most suitable choice, and why the others are less appropriate:

D. Groundedness Evaluation Metrics:

Relevance: Groundedness metrics directly measure how well the model's responses are grounded in the retrieved content from Azure AI Search. A decline in these metrics post-content update would indicate that the new content is not being effectively utilized or is misleading the model, leading to less accurate responses.

Actionability: Reviewing these metrics provides clear insights into whether the issue lies in the content update itself or in how the model interprets this content, guiding targeted adjustments.

A. Indexer Status and Failure History:

Less Suitable Because: While important for overall system health, indexer status primarily indicates if the indexing process is functioning correctly. Failure history might show if content wasn't indexed properly, but it doesn't directly correlate with the model's ability to generate accurate responses from the content.

Why Not Best: Doesn't provide insight into the model's usage of the content for response accuracy.

B. Latency Breakdown Traces:

Less Suitable Because: Latency issues might affect user experience but do not directly impact the accuracy of the model's responses. Slow performance could be a separate issue but doesn't explain the decrease in response accuracy.

Why Not Best: Focuses on performance rather than content-model interaction accuracy.

C. Prediction Drift Metrics:

Less Suitable Because: While prediction drift metrics are crucial for monitoring model performance over time, they are more generalized. They can indicate a decline in model accuracy but do not specifically point to the influence of the retrieved content as the cause.

Why Not Best: Less targeted than groundedness metrics for identifying content-related issues.

Conclusion: Given the need to isolate whether the retrieved content update is negatively impacting the model's response accuracy, **D. Groundedness Evaluation Metrics** is the most direct and relevant observability signal to review. It offers the clearest path to understanding the interaction between the new content and the model's performance.

References

[Microsoft Documentation: Azure AI Search - Troubleshooting](#) (Although not directly linked to "Groundedness Evaluation Metrics", this resource provides insight into troubleshooting content indexing issues, which can be a precursor to evaluating groundedness.)

[Microsoft Learn: Monitor and troubleshoot Azure AI models](#) (While broad, this learning module touches on model monitoring aspects that can be related to understanding prediction drift and, by extension, the importance of specific metrics in model performance analysis.)

Thank you

Thank you for being so interested in the premium exam material.
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If you have any feedback or thoughts on the bumps, I would love to hear them.
Your insights can help me improve our writing and better understand our readers.

Best of Luck

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Keep your head up, stay positive, and go show that exam what you're made of!

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